

Lecture 12
Parametrised Quantum Circuits
Variational Ansätze, Born Machines and QAOA

Deep Learning in Classical and Quantum Physics

Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków

1

Many-Body States and Approximations



JAGIELLONIAN UNIVERSITY
IN KRAKOW

The Exponential Wall

A state of N spins-1/2 is a vector of 2^N complex amplitudes, one for every bit string $b_0 b_1 \ldots b_{N-1}$. In double precision every amplitude takes 16 bytes.

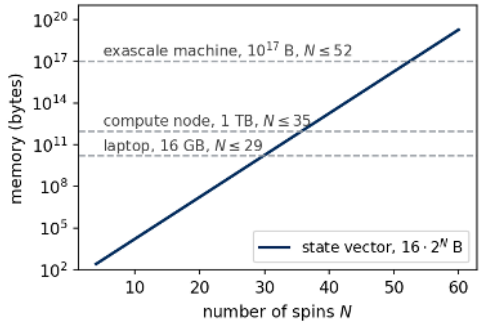
spins N	10	30	50
memory of one state, $16 \cdot 2^N$ bytes	16 kB	17 GB	18 PB

Ground states, excited states and the evolution in time all need repeated products $\hat{\mathcal{H}}|\psi\rangle$ of the Hamiltonian with such vectors. The cost of an exact calculation therefore doubles with every added spin, and exact diagonalisation stops at a few dozen spins.

Approximations

Decades of many-body physics produced trial states that store far fewer numbers. Each family captures some physics and misses some. This lecture starts with three classical families and ends with trial states prepared by a quantum register.

Memory of a State Vector



our simulation

Memory $16 \cdot 2^N$ bytes of one state vector against N, with a laptop, a compute node and an exascale machine as dashed lines.

Each added spin doubles the memory, so even the largest machine stores the state of only about fifty spins.

The Transverse-Field Ising Chain

Our running model is an open chain of N qubits,

$$\hat{\mathcal{H}} = \underbrace{-J \sum_{i=0}^{N-2} \hat{Z}_i \hat{Z}_{i+1}}_{\text{neighbours aligned along } z} \qquad \underbrace{- h \sum_{i=0}^{N-1} \hat{X}_i}_{\text{each spin along } +x} \quad , \qquad J = 1.$$

- The two terms do not commute, so the ground state is a superposition of many bit strings.
- For $h < J$ the chain is ordered along z, for $h > J$ it is a paramagnet. In a long chain the quantum phase transition lies at $h = J$.
- Two product states give quick upper bounds on the ground-state energy E_0. In $|0 \ldots 0\rangle$ the energy is $-J(N-1)$, in $|+ \ldots +\rangle$ it is $-hN$.

At $h = J$ and $N = 6$ the exact values are $E_0 = -7.2962$ and $E_1 = -6.8141$, while the two product states give -5 and -6.

The Variational Principle

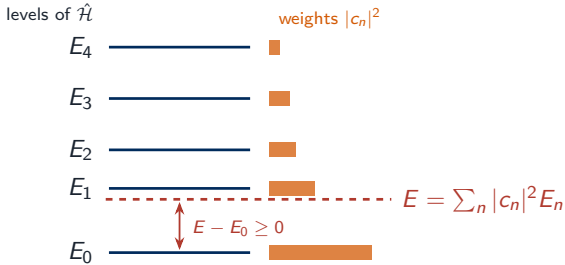
The eigenvalues $E_0 \leq E_1 \leq \ldots$ of $\hat{\mathcal{H}}$ have orthonormal eigenvectors $|E_m\rangle$, so a normalised trial state is $|\psi\rangle = \sum_m c_m |E_m\rangle$ with $\sum_m |c_m|^2 = 1$. Its energy is

$$E = \langle\psi|\hat{\mathcal{H}}|\psi\rangle = \sum_m |c_m|^2 \underbrace{E_m}_{\geq E_0} \geq E_0 \underbrace{\sum_m |c_m|^2}_{=1} = E_0.$$

Every trial state gives an upper bound on the ground-state energy, as the two product states of the Ising chain do. Minimising E over a family of trial states $|\psi(\boldsymbol{\theta})\rangle$, the **ansatz**, gives the best bound that the family allows (Rayleigh–Ritz method).

Every approximation in this part is such a family. Mean field, tensor networks and variational Monte Carlo choose different families of trial states that a classical computer can store.

Trial Energy as an Average of Levels



trial state $|\psi\rangle = \sum_n c_n |E_n\rangle$

its energy is the average of the levels with weights $|c_n|^2$, $\sum_n |c_n|^2 = 1$

an average never lies below the lowest level, so $E \geq E_0$

The trial energy is the average of the energy levels with the weights $|c_n|^2$. It reaches E_0 only when all weight sits on the ground level.

Bosons in Second Quantisation

Mean-field theory is clearest for identical bosons in a trap, e.g. the atoms of a Bose–Einstein condensate. A single boson has orthonormal wave functions $\phi_i(x)$, its *modes*. The operator $\hat{a}_i^\dagger$ adds a boson to mode i and $\hat{a}_i$ removes one,

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij}, \qquad [\hat{a}_i, \hat{a}_j] = 0, \qquad \hat{\psi}(x) = \sum_i \phi_i(x)\, \hat{a}_i .$$

The field operator $\hat{\psi}(x)$ removes one boson at position x from whichever mode it occupies. For bosons of mass m in a trap $V(x)$, interacting with strength g at contact, the Hamiltonian in one dimension is

$$\hat{\mathcal{H}}_B = \underbrace{\int dx\, \hat{\psi}^\dagger(x) \Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x) \Big] \hat{\psi}(x)}_{\text{kinetic and trap energy of every boson}} + \underbrace{\frac{g}{2} \int dx\, \hat{\psi}^\dagger(x)\hat{\psi}^\dagger(x)\hat{\psi}(x)\hat{\psi}(x)}_{\text{every pair at the same point}} .$$

All Bosons in One Mode

The mean-field ansatz puts all N bosons into one normalised mode ϕ, $\int dx\,|\phi|^2 = 1$,

$$|\Psi_N\rangle = \frac{(\hat{a}_\phi^\dagger)^N}{\sqrt{N!}}\,|\mathrm{vac}\rangle, \qquad \Psi_N(x_1,\ldots,x_N) = \phi(x_1)\cdots\phi(x_N).$$

Only the component $\phi(x)\hat{a}_\phi$ of $\hat{\psi}(x)$ acts on this state, $\hat{\psi}(x)|\Psi_N\rangle = \sqrt{N}\,\phi(x)|\Psi_{N-1}\rangle$, so

$$\langle\hat{\psi}^\dagger(x)\hat{\psi}(x')\rangle = \underbrace{N\,\phi^*(x)\phi(x')}_{\text{one boson removed and replaced}}, \qquad \langle\hat{\psi}^\dagger\hat{\psi}^\dagger\hat{\psi}\hat{\psi}\rangle(x) = \underbrace{N(N-1)\,|\phi(x)|^4}_{\text{ordered pairs of bosons}}.$$

Inserting both into $\hat{\mathcal{H}}_B$ turns the energy into a functional of the single mode,

$$E[\phi] = N\int dx\,\phi^*\Big[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V\Big]\phi + \frac{gN(N-1)}{2}\int dx\,|\phi|^4.$$

The Gross–Pitaevskii Equation

We minimise $E[\phi]$ at fixed normalisation. A Lagrange multiplier μ enforces the constraint, and we treat ϕ and ϕ^* as independent variables,

$$\frac{\delta}{\delta\phi^*(x)}\Big\{E[\phi] - N\mu\Big(\int dx\,|\phi|^2 - 1\Big)\Big\} = 0, \qquad \frac{\delta}{\delta\phi^*}\,(\phi^*)^2\phi^2 = 2|\phi|^2\phi.$$

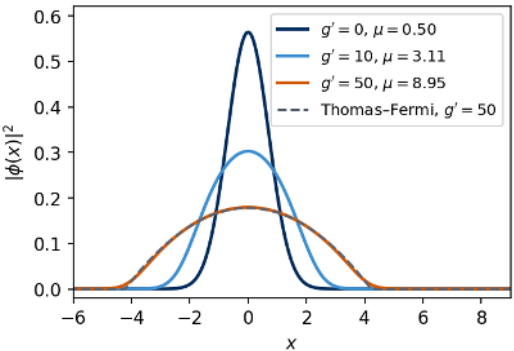
Dividing by N gives the Gross–Pitaevskii equation

$$\Big[\underbrace{-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)}_{\text{one boson}} + \underbrace{g(N-1)\,|\phi(x)|^2}_{\text{mean field of the other } N-1}\Big]\phi(x) = \underbrace{\mu}_{\text{chemical potential}}\,\phi(x).$$

- It is a Schrödinger equation whose potential depends on its own solution, so it is nonlinear and solved self-consistently.
- μ is the energy needed to add one boson to the condensate.

E. P. Gross, Nuovo Cimento **20**, 454 (1961), doi:10.1007/BF02731494 L. P. Pitaevskii, Sov. Phys. JETP **13**, 451 (1961), jetp.ras.ru

A Condensate in a Harmonic Trap



our simulation

We solve the equation for $V(x) = x^2/2$ (units $\hbar = m = \omega = 1$) by evolution in imaginary time, which damps every excited component, at the interaction strengths $g' = g(N-1) = 0$, 10 and 50.

Without interactions the condensate is the Gaussian oscillator ground state with $\mu = 0.500$. Repulsion spreads the cloud and raises μ to 3.11 and 8.95. At $g' = 50$ the density is close to the Thomas–Fermi parabola $|\phi|^2 = (\mu - x^2/2)/g'$ with $\mu_{\mathrm{TF}} = 8.89$, which neglects the kinetic energy.

Mean Field for the Ising Chain

The same logic for spins puts every spin into the same single-qubit state,

$$|\psi_{\mathrm{MF}}(\theta)\rangle = |\phi(\theta)\rangle^{\otimes N}, \qquad |\phi(\theta)\rangle = \cos\tfrac{\theta}{2}|0\rangle + \sin\tfrac{\theta}{2}|1\rangle = \hat{R}_y(\theta)|0\rangle,$$

with $\langle \hat{Z}_i \rangle = \cos\theta$ and $\langle \hat{X}_i \rangle = \sin\theta$. One angle replaces 2^N amplitudes. In a product state the correlator factorises, $\langle \hat{Z}_i \hat{Z}_{i+1} \rangle = \langle \hat{Z}_i \rangle \langle \hat{Z}_{i+1} \rangle$, so

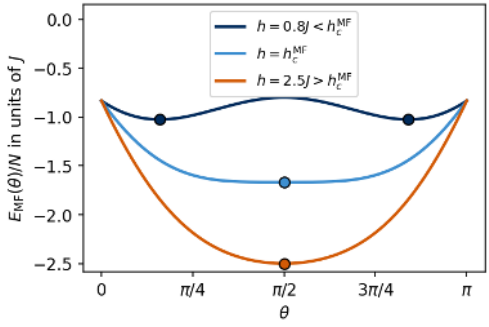
$$E_{\mathrm{MF}}(\theta) = \underbrace{-J(N-1)\cos^2\theta}_{N-1 \text{ bonds},\ \cos\theta\cdot\cos\theta} \underbrace{- hN\sin\theta}_{N \text{ spins}}, \qquad \frac{E_{\mathrm{MF}}}{N} \xrightarrow{N\to\infty} -J\cos^2\theta - h\sin\theta.$$

The best angle makes the derivative vanish, the self-consistent equation of mean-field theory,

$$\frac{dE_{\mathrm{MF}}}{d\theta} = \cos\theta\left[2J(N-1)\sin\theta - hN\right] = 0.$$

Each spin points along the total field made of h along x and the field $2J\langle \hat{Z} \rangle$ of its neighbours along z (factor $(N-1)/N$ on the open chain).

The Mean-Field Energy Landscape



our simulation

$E_{\mathrm{MF}}(\theta)/N$ for the open chain below, at and above h_c^{MF}. Dots mark the minima.

Below h_c^{MF} the energy has two symmetric minima, the two magnetised solutions, which merge at $\theta = \pi/2$ above it.

The Mean-Field Transition

The minimum of $E_{\mathrm{MF}}(\theta)$ has two branches,

$$h < h_c^{\mathrm{MF}}: \quad \sin\theta^* = \frac{h}{h_c^{\mathrm{MF}}}, \quad E_{\mathrm{MF}} = -J(N-1) - \frac{h^2 N^2}{4J(N-1)},$$

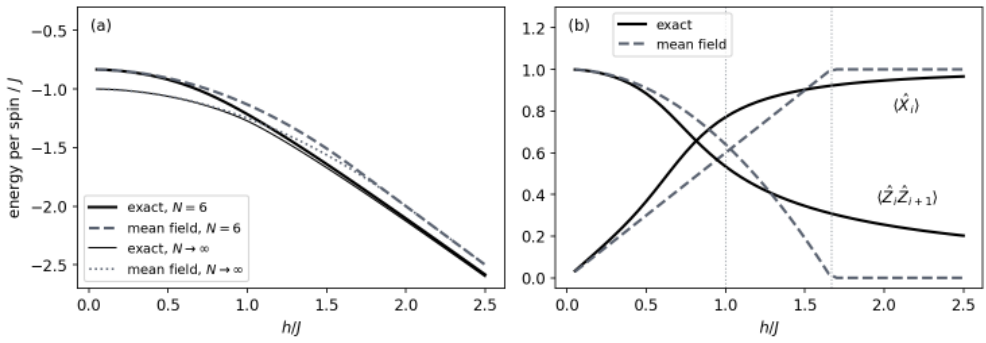
$$h \geq h_c^{\mathrm{MF}}: \quad \theta^* = \frac{\pi}{2}, \quad E_{\mathrm{MF}} = -hN,$$

with $h_c^{\mathrm{MF}} = 2J(N-1)/N$, which is $1.667J$ for $N = 6$ and $2J$ for the infinite chain.

- Below h_c^{MF} the spins tilt and keep a magnetisation $\pm\cos\theta^*$, two solutions related by $\hat{Z} \to -\hat{Z}$.
- The exact transition of the infinite chain lies at $h_c = J$. Mean field puts it at $2J$ and predicts order in $J < h < 2J$, where there is none.
- The exact energy per spin of the infinite chain follows from a mapping to free fermions, $e_\infty(h) = -\frac{1}{\pi}\int_0^\pi dk\,\sqrt{J^2 + h^2 - 2Jh\cos k}$. At $h = J$ it is $-1.2732J$, whereas mean field gives $-1.2500J$.

P. Pfeuty, Ann. Phys. **57**, 79 (1970), doi:10.1016/0003-4916(70)90270-8

Mean Field Against the Exact Chain



our simulation

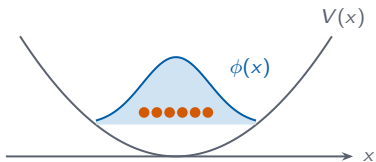
(a) Energy per spin, mean field and exact, for $N = 6$ and the infinite chain. (b) $\langle \hat{Z}_i \hat{Z}_{i+1} \rangle$ and $\langle \hat{X}_i \rangle$ for $N = 6$. Dotted lines mark h_c^{MF}.

Limits of Mean Field

- Every mean-field energy lies above its exact counterpart, most between the exact critical point and the mean-field transition. At $h = J$ and $N = 6$ the error is 0.068 of $|E_0|$.
- Far from the critical point, at small and at large h, the exact ground state is close to a product state and mean field becomes accurate.
- Above h_c^{MF} the mean-field correlator vanishes, whereas the exact one stays positive, because quantum fluctuations correlate neighbours without a magnetisation.
- A product state has zero entanglement by construction. In circuit language it is one $\hat{R}_y(\theta)$ on every qubit and no entangling gate.

Product Ansätze for Bosons and Spins

(a) bosons in a trap, one mode for all



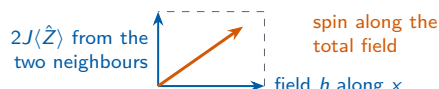
$$\Psi_N(x_1, \ldots, x_N) = \phi(x_1)\,\phi(x_2)\cdots\phi(x_N)$$

Hartree mean field $\Rightarrow$ Gross–Pitaevskii equation

(b) spins on a chain, one state per site



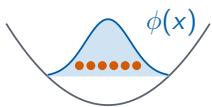
$$|\psi_{\mathrm{MF}}\rangle = |\phi(\theta)\rangle^{\otimes N}, \quad \langle \hat{Z}_i \rangle = \cos\theta, \quad \langle \hat{X}_i \rangle = \sin\theta$$



In both cases every particle or site carries its own copy of one single-particle state. The energy becomes a functional of that state, and its minimum obeys a nonlinear self-consistent equation.

Particle and Site Product States

(a) all bosons in one mode (Hartree)

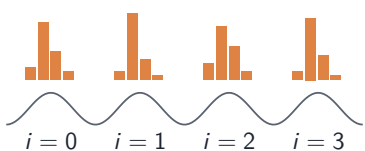


$|\Psi_N\rangle = \frac{1}{\sqrt{N!}}(\hat{a}_\phi^\dagger)^N|\mathrm{vac}\rangle$

product over particles,

$\Psi_N = \phi(x_1)\cdots\phi(x_N)$

(b) one state per lattice site (Gutzwiller)



$|\Psi\rangle = \bigotimes_i \sum_n f_n^{(i)}|n\rangle_i$

bars show $|f_n^{(i)}|^2$, $n = 0, \ldots, 3$

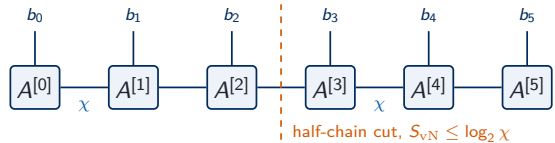
product over sites

Two Families of Mean Field

Hartree. The Gross–Pitaevskii state factorises over particles, all in one mode, and each particle moves in the average potential of all others. **Site-factorised.** The spin state $\bigotimes_i |\phi_i\rangle$ factorises over sites, a spin coherent state in the molecular field of Weiss. For lattice bosons the Gutzwiller ansatz $\prod_i \sum_n f_n^{(i)} |n\rangle_i$ describes the superfluid–Mott insulator transition of the Bose–Hubbard model. Neither family contains entanglement between its factors.

P. Weiss, J. Phys. Theor. Appl. **6**, 661 (1907), doi:10.1051/jphystap:019070060066100 M. C. Gutzwiller, PRL **10**, 159 (1963), doi:10.1103/PhysRevLett.10.159 M. P. A. Fisher *et al.*, PRB **40**, 546 (1989), doi:10.1103/PhysRevB.40.546 D. S. Rokhsar and B. G. Kotliar, PRB **44**, 10328 (1991), doi:10.1103/PhysRevB.44.10328

Tensor Networks



$\psi(b_0, \ldots, b_{N-1}) = A^{[0]}_{b_0} A^{[1]}_{b_1} \cdots A^{[N-1]}_{b_{N-1}}$, each $A^{[i]}_{b_i}$ a matrix of size at most $\chi \times \chi$

- A **matrix product state** (MPS) writes each amplitude as a product of N matrices of size at most $\chi \times \chi$, the *bond dimension*. It stores about $2N\chi^2$ numbers.
- The entanglement entropy across any cut is at most $\log_2 \chi$. Ground states of gapped one-dimensional chains with local interactions obey an *area law*, a bounded entropy as N grows, so a modest χ suffices.
- The density-matrix renormalisation group (DMRG) optimises an MPS variationally.
- The required χ grows exponentially with the entanglement, which grows with the width of two-dimensional systems and linearly in time after a quench.

S. R. White, PRL **69**, 2863 (1992), doi:10.1103/PhysRevLett.69.2863

Monte Carlo Methods

Monte Carlo methods estimate expectation values from random samples and never store the full state.

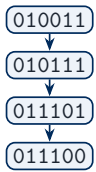
- **Quantum Monte Carlo** maps the quantum problem onto sampling classical configurations with non-negative weights.
- For fermions and frustrated magnets the weights take both signs. The statistical error then grows exponentially with the system size and the inverse temperature. This **sign problem** is NP-hard in general.
- **Variational Monte Carlo** samples bit strings $\mathbf{b}$ from $|\psi(\mathbf{b})|^2$ of a trial wave function computed by a classical formula. A neural network can be this formula, with weights trained to minimise the energy.

Mean field excludes entanglement, tensor networks limit it, and Monte Carlo is limited by the sign structure of the amplitudes or by the form of the trial wave function.

M. Troyer and U.-J. Wiese, PRL **94**, 170201 (2005), doi:10.1103/PhysRevLett.94.170201 G. Carleo and M. Troyer, Science **355**, 602 (2017), doi:10.1126/science.aag2302

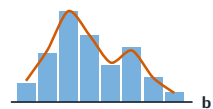
Sampling Configurations

(a) a chain of configurations

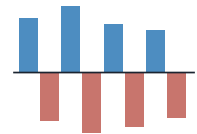


flip a bit, accept with
$\min\,(1, |\psi(\mathbf{b}')|^2/|\psi(\mathbf{b})|^2)$

(b) visits follow $|\psi(\mathbf{b})|^2$



(c) sign problem



weights of both signs cancel,
the error grows exponentially with N

(a) A Markov chain proposes bit flips and accepts them with the ratio of the weights. (b) Its visits follow $|\psi(\mathbf{b})|^2$, so averages over visited configurations estimate expectation values. (c) When the weights take both signs, they cancel and the statistical error grows exponentially with N.

Trial States on a Quantum Register

Experiments with trapped ions, neutral atoms in optical tweezers and superconducting circuits control tens to hundreds of qubits individually. Every qubit is prepared in $|0\rangle$, rotated by laser or microwave pulses, coupled to its neighbours and measured. Such a register holds all 2^N amplitudes physically.

Parametrised quantum circuit

- architecture set by the hardware
- gate angles are pulse areas $\theta = \Omega t$ and interaction times $\theta = Jt$
- angles optimised by a classical computer

A fully quantum ansatz

The trial state exists only on the register. Its amplitudes are never written down, and the device returns bit strings, from which energies are estimated.

The rest of the lecture develops this idea for ground states, then reuses the same circuits as probability models and as optimisers of binary problems.

2

Parametrised Circuits and the Hybrid Loop



JAGIELLONIAN UNIVERSITY
IN KRAKOW

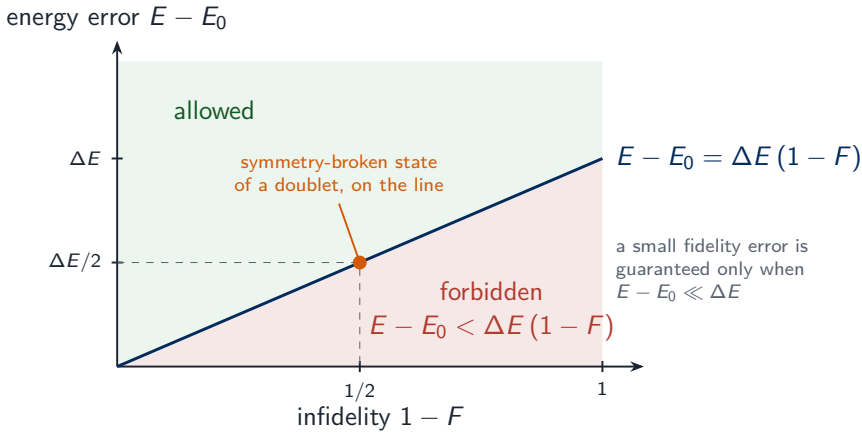
Energy Error and Fidelity

The energy error bounds how far the trial state is from the ground state. With the fidelity $F = |\langle E_0|\psi\rangle|^2 = |c_0|^2$ and the gap $\Delta E = E_1 - E_0$,

$$E - E_0 = \sum_{m\geq 1} |c_m|^2 \underbrace{(E_m - E_0)}_{\geq \Delta E} \;\geq\; \Delta E \underbrace{\sum_{m\geq 1} |c_m|^2}_{=1-F} = \Delta E\,(1-F).$$

A small energy error guarantees a fidelity close to one only when it is small compared with the gap, so a circuit ansatz has to be judged by both quantities.

The Fidelity Bound



The Parametrised Circuit

The register starts in $|0\ldots0\rangle$, and a circuit with n adjustable angles $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_n)$ prepares the trial state,

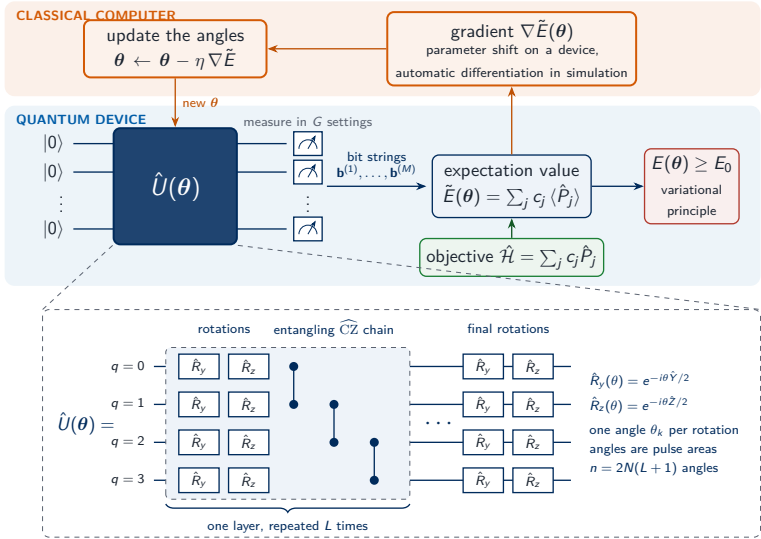
$$|\psi(\boldsymbol{\theta})\rangle = \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle, \qquad \hat{U}(\boldsymbol{\theta}) = \hat{W}_n\, e^{-i\theta_n \hat{P}_n/2} \cdots \hat{W}_1\, e^{-i\theta_1 \hat{P}_1/2}.$$

- Each $\hat{P}_k$ is a Pauli string (a product of $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ on the qubits), and each $\hat{W}_k$ is a fixed gate without an angle, often the identity.
- $e^{-i\theta\hat{P}/2} = \hat{R}_P(\theta)$ is a physical operation. A one-qubit rotation is a pulse of Rabi frequency Ω and duration t with $\theta = \Omega t$, a two-qubit rotation an interaction of strength J switched on for t, with $\theta = Jt$.
- The energy $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{\mathcal{H}}|\psi(\boldsymbol{\theta})\rangle$ is a function of n real numbers, never below E_0.

A quantum processor prepares the trial state, and the hybrid scheme is the **variational quantum eigensolver** (VQE).

A. Peruzzo *et al.*, Nat. Commun. **5**, 4213 (2014), doi:10.1038/ncomms5213 J. R. McClean *et al.*, New J. Phys. **18**, 023023 (2016), doi:10.1088/1367-2630/18/2/023023

The Variational Loop and the Ansatz



The Hybrid Quantum–Classical Loop

1. The classical computer sends the angles $\boldsymbol{\theta}$ to the device.
2. The device prepares $|\psi(\boldsymbol{\theta})\rangle$, rotates the qubits into each measurement setting and records M bit strings per setting.
3. The classical computer forms $\tilde{E}(\boldsymbol{\theta})$ and, from runs at shifted angles, an estimate of the gradient.
4. It updates $\boldsymbol{\theta}$ to lower the energy and repeats until the energy stops decreasing.

In our simulations a state-vector calculation plays the device, so every estimate can be compared with its exact value.

3

The Energy from Bit Strings



JAGIELLONIAN UNIVERSITY
IN KRAKOW

The Energy as a Sum of Pauli Strings

Any Hamiltonian of qubits is a weighted sum of Pauli strings, $\hat{\mathcal{H}} = \sum_j c_j \hat{P}_j$, so the energy is a sum of expectation values,

$$E(\boldsymbol{\theta}) = \sum_j c_j \langle \hat{P}_j \rangle.$$

The Ising chain has $N-1$ strings $\hat{Z}_i \hat{Z}_{i+1}$ with $c_j = -J$ and N strings $\hat{X}_i$ with $c_j = -h$.

Setting Z, all qubits measured

Bit string **b** gives $\hat{Z}_i \to (-1)^{b_i}$ and

$$\hat{Z}_i \hat{Z}_{i+1} \to (-1)^{b_i + b_{i+1}},$$

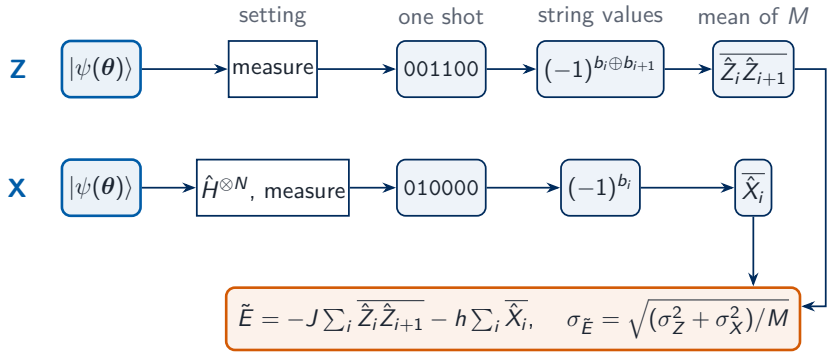
the parity of the two bits.

Setting X, Hadamard gates first

$\hat{H}\hat{X}\hat{H} = \hat{Z}$, so after a Hadamard gate $\hat{H}$ on every qubit the bit b_i gives $\hat{X}_i \to (-1)^{b_i}$.

Two measurement settings are enough for the Ising chain. The XXZ chain, $\hat{X}\hat{X} + \hat{Y}\hat{Y} + \Delta \hat{Z}\hat{Z}$, needs three, since $\hat{Y}$ is measured after $\hat{S}^\dagger$ and $\hat{H}$. ($\hat{H}$ is the Hadamard gate, $\hat{\mathcal{H}}$ the Hamiltonian.)

Two Measurement Settings



Each shot of a setting gives one bit string, from which every Pauli string of that setting takes the value ± 1. Averages over M shots give the expectation values, and their weighted sum gives $\tilde{E}$. Commuting strings diagonal in the same basis share one setting, so the Ising chain needs two.

The Estimator and Its Error

Each shot of setting $g \in \{Z, X\}$ gives one value of its part of the energy,

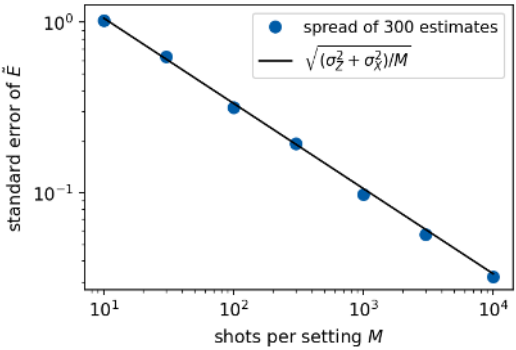
$$e_Z(\mathbf{b}) = -J \sum_{i=0}^{N-2} (-1)^{b_i + b_{i+1}}, \qquad e_X(\mathbf{b}) = -h \sum_{i=0}^{N-1} (-1)^{b_i},$$

and M shots per setting give the estimate and its standard error,

$$\tilde{E} = \underbrace{\frac{1}{M} \sum_{s=1}^{M} e_Z(\mathbf{b}^{(s)})}_{\text{sample mean, setting } Z} + \underbrace{\frac{1}{M} \sum_{s=1}^{M} e_X(\mathbf{b}'^{(s)})}_{\text{sample mean, setting } X}, \qquad \sigma_{\tilde{E}} = \sqrt{\frac{\sigma_Z^2 + \sigma_X^2}{M}},$$

where σ_g^2 is the variance of a single shot of setting g. For one Pauli string with values ± 1 it equals $1 - \langle \hat{P} \rangle^2$. The mean of $\tilde{E}$ over repetitions is the exact energy, and the two independent settings add their variances.

Shot Noise of the Energy



our simulation Spread of 300 repeated estimates $\tilde{E}$ against the shots M per setting, with the prediction $\sigma_{\tilde{E}}$, at fixed random angles.

The spread follows $M^{-1/2}$ over three orders of magnitude. With $M = 1000$ shots per setting the estimate is -0.172 ± 0.107 at a state of exact energy -0.186.

A tenfold smaller error bar costs a hundred times more shots. A standard error of $0.01J$ at these angles needs about 1.1×10^5 shots per setting, and the single-shot variances, hence the shot count, change from state to state.

4

Gradients

JAGIELLONIAN UNIVERSITY
IN KRAKOW

The Energy Along One Angle

We split the circuit at the rotation that carries θ_k, $\hat{U}(\boldsymbol{\theta}) = \hat{A}\, e^{-i\theta_k \hat{P}_k/2}\, \hat{B}$, where $\hat{A}$ and $\hat{B}$ do not depend on θ_k. With $|\phi\rangle = \hat{B}|0\ldots0\rangle$ and $\hat{O} = \hat{A}^\dagger \hat{\mathcal{H}} \hat{A}$, and $e^{-i\theta\hat{P}/2} = \cos\frac{\theta}{2}\hat{I} - i\sin\frac{\theta}{2}\hat{P}$ from $\hat{P}^2 = \hat{I}$,

$$\begin{aligned} E(\theta_k) &= \langle\phi| e^{i\theta_k \hat{P}_k/2}\, \hat{O}\, e^{-i\theta_k \hat{P}_k/2} |\phi\rangle \\ &= \underbrace{\cos^2\tfrac{\theta_k}{2}}_{(1+\cos\theta_k)/2}\, \langle\hat{O}\rangle + \underbrace{\sin^2\tfrac{\theta_k}{2}}_{(1-\cos\theta_k)/2}\, \langle\hat{P}_k\hat{O}\hat{P}_k\rangle + \underbrace{i\sin\tfrac{\theta_k}{2}\cos\tfrac{\theta_k}{2}}_{(i/2)\sin\theta_k} \langle[\hat{P}_k, \hat{O}]\rangle. \end{aligned}$$

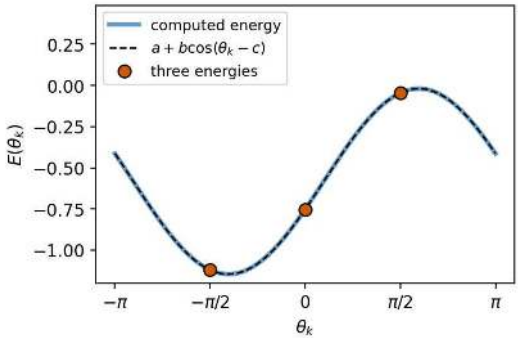
Collecting terms gives a shifted cosine of period 2π,

$$E(\theta_k) = a + u_k\cos\theta_k + v_k\sin\theta_k = a + b\cos(\theta_k - c),$$

$$a = \tfrac{1}{2}(\langle\hat{O}\rangle + \langle\hat{P}_k\hat{O}\hat{P}_k\rangle), \quad u_k = \tfrac{1}{2}(\langle\hat{O}\rangle - \langle\hat{P}_k\hat{O}\hat{P}_k\rangle), \quad v_k = \tfrac{i}{2}\langle[\hat{P}_k, \hat{O}]\rangle.$$

Only $\hat{P}_k^2 = \hat{I}$ was used, so this holds for every rotation of the circuit and every Hamiltonian. The constants depend on all other angles.

The Cosine Checked Numerically



our simulation Energy against one $\hat{R}_y$ angle with all others fixed, computed at 201 angles and as the cosine through the three points $\theta_k = 0, \pm\pi/2$.

Three energies fix the three constants, $E(0) = a + u_k$ and $E(\pm\pi/2) = a \pm v_k$. The cosine through them, with $a = -0.582$, $b = 0.563$, $c = 1.875$, coincides with the computed energy at all 201 angles to numerical precision.

Along any single angle the landscape has one minimum and one maximum. The difficulty of the optimisation comes from the n angles together.

The Parameter-Shift Rule

Differentiating the cosine gives $\partial E/\partial\theta_k = -u_k \sin\theta_k + v_k \cos\theta_k$. With $\cos(\theta \pm \frac{\pi}{2}) = \mp \sin\theta$ and $\sin(\theta \pm \frac{\pi}{2}) = \pm \cos\theta$, the same combination appears in the difference of two energies at angles shifted by $\pm\pi/2$,

$$E(\theta_k + \tfrac{\pi}{2}) - E(\theta_k - \tfrac{\pi}{2}) = \underbrace{u_k\big(-\sin\theta_k - \sin\theta_k\big)}_{\text{from } u_k \cos\theta_k} + \underbrace{v_k\big(\cos\theta_k + \cos\theta_k\big)}_{\text{from } v_k \sin\theta_k} = 2\,\frac{\partial E}{\partial \theta_k}.$$

Hence, with $\mathbf{e}_k$ the unit vector along angle k,

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\boldsymbol{\theta} + \frac{\pi}{2}\mathbf{e}_k) - E(\boldsymbol{\theta} - \frac{\pi}{2}\mathbf{e}_k)}{2}.$$

The shift is finite and the rule is exact, with both energies from the same device and circuit. For a pulse of Rabi frequency Ω the shifted circuits differ only in the pulse duration, $t_k \pm \pi/(2\Omega)$.

K. Mitarai *et al.*, PRA **98**, 032309 (2018), doi:10.1103/PhysRevA.98.032309 M. Schuld *et al.*, PRA **99**, 032331 (2019), doi:10.1103/PhysRevA.99.032331

Shared Angles and the Chain Rule

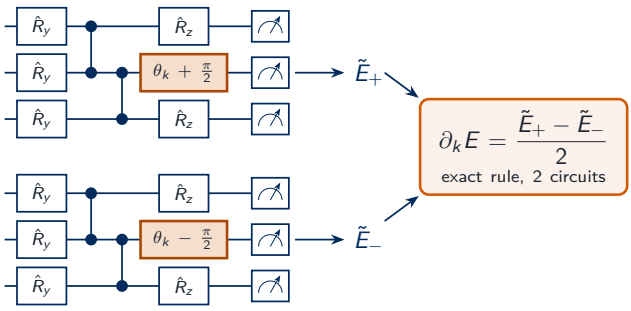
Many ansätze use one angle φ in several gates. If gate g rotates by $\theta_g = c_g \varphi$ with a constant c_g, the chain rule gives

$$\frac{\partial E}{\partial \varphi} = \sum_g c_g \underbrace{\frac{\partial E}{\partial \theta_g}}_{\text{two-point rule, gate } g \text{ shifted alone}} .$$

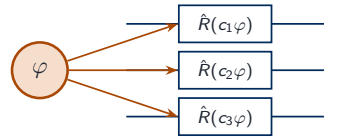
- The hardware cost counts the parametrised Pauli rotations, the occurrences of angles in gates, two circuits each. It does not count the free angles.
- The mean-field state $\hat{R}_y(\theta)^{\otimes N}|0\ldots0\rangle$ is an example, with θ in N gates and $c_g = 1$. The rule reproduces $dE_{\mathrm{MF}}/d\theta$ of the mean-field energy.

Shifted Circuits and Shared Angles

(a) one angle, two shifted circuits



(b) one angle in several gates



$$\frac{\partial E}{\partial \varphi} = \sum_g c_g \frac{\partial E}{\partial \theta_g}$$

each term by its own shift pair, two circuits per gate occurrence

(a) The same circuit runs twice with one angle shifted by $\pm\pi/2$, and half the difference of the two estimated energies is the derivative. (b) A shared angle needs one such pair for each gate it enters.

Automatic Differentiation and the Cost of a Gradient

Simulation

Reverse-mode automatic differentiation applies the chain rule to every operation of the program that computes E, backwards through the stored intermediate states. `jax.grad` returns the whole gradient for the cost of a few energy evaluations, for any n. It agrees with the parameter-shift rule to numerical precision.

Device

The device has no intermediate states, and the parameter-shift rule is the exact method,

$$\text{shots per gradient} = 2\, n_{\mathrm{rot}}\, G\, M,$$

with n_{rot} the number of parametrised Pauli rotations (gate occurrences of the angles), G settings and M shots per setting.

The six-qubit HEA with $n_{\mathrm{rot}} = n = 60$, $G = 2$ and $M = 1000$ need 240000 shots for one gradient and about 1.4×10^8 for 600 training steps.

Finite differences need the same $2n_{\mathrm{rot}}$ circuits, with shot noise growing as one over the step.

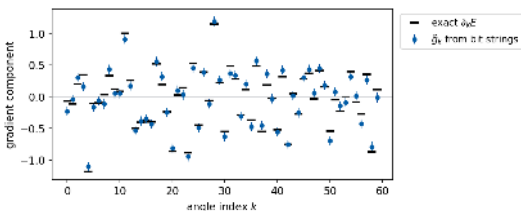
SPSA shifts all angles along a random ± 1 direction, two energies per step, with a noisy estimate.

We update the angles with Adam, a gradient method with step sizes adapted to each angle.

S. Linnainmaa, BIT **16**, 146 (1976), doi:10.1007/BF01931367 J. C. Spall, IEEE Trans. Autom. Control **37**, 332 (1992), doi:10.1109/9.119632 D. P. Kingma and J. Ba, arXiv:1412.6980

A Gradient from Bit Strings

under the hood



our simulation The 60 gradient components estimated from $M = 1000$ shots per setting and per shifted circuit, against their exact values.

Replacing both energies of the shift rule by estimates gives a component $\tilde{g}_k = (\tilde{E}_{k,+} - \tilde{E}_{k,-})/2$ with the standard error

$$\sigma_{\tilde{g}_k} \approx \sqrt{\frac{\sigma^2}{2M}}, \qquad \sigma^2 = \sigma_Z^2 + \sigma_X^2.$$

The estimates scatter around the exact values within their error bars. Components smaller than about twice the error bar cannot be told apart from zero, and halving the error bar costs four times the shots.

5
Circuit Ansätze

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Requirements on an Ansatz

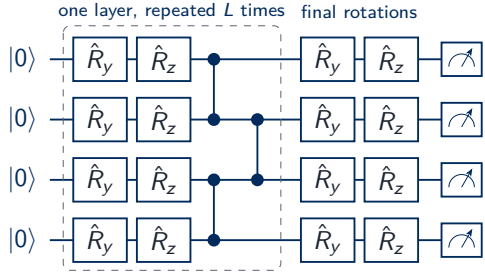
The family of states that one circuit layout can prepare is the ansatz. It limits the quality of every trained state, so it has to be chosen with the physics of the target in mind.

1. **Expressibility.** The family contains a good approximation of the target state.
2. **Trainability.** Gradient descent finds good angles with a feasible number of shots.
3. **Hardware.** The circuit uses the native gates and the qubit connectivity of the device, and is shallow, because every gate adds noise.
4. **Symmetry.** Ideally the circuit conserves the symmetries of $\hat{\mathcal{H}}$, so it stays in the symmetry sector of the target.

In quantum chemistry the standard choice is the unitary coupled-cluster ansatz, which builds the trial state from excitations of electrons out of a reference configuration. For spin chains we compare three ansätze that weigh these demands differently.

J. Romero *et al.*, Quantum Sci. Technol. **4**, 014008 (2019), doi:10.1088/2058-9565/aad3e4

The Hardware-Efficient Ansatz



independent angle in every rotation, $n = 2N(L+1)$, $\widehat{\mathrm{CZ}}$ chain entangles neighbours

- Start $|0\ldots0\rangle$, L layers of independent $\hat{R}_y\hat{R}_z$ rotations and a $\widehat{\mathrm{CZ}}$ chain, final rotations, $n = 2N(L+1)$.
- The first block alone contains every product state, the $\widehat{\mathrm{CZ}}$ gates create entanglement, and no layer conserves a symmetry of $\hat{\mathcal{H}}$.

A. Kandala *et al.*, Nature **549**, 242 (2017), doi:10.1038/nature23879

The Hamiltonian-Variational Ansatz

The HVA builds every layer from the terms of $\hat{\mathcal{H}}$. For the Ising chain it starts in $|+\rangle^{\otimes N}$, the ground state of the field term,

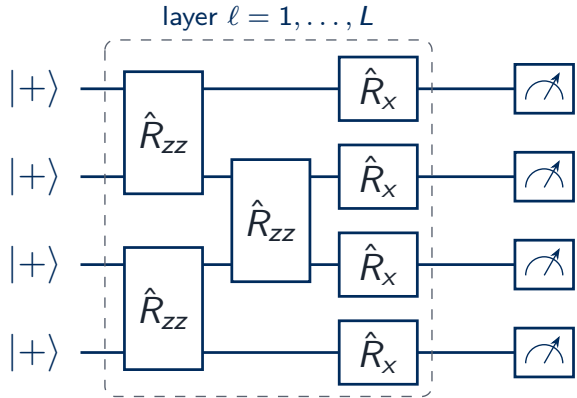
$$|\psi_{\mathrm{HVA}}\rangle = \prod_{\ell=1}^{L} \underbrace{e^{-i\beta_\ell \sum_i \hat{X}_i}}_{\hat{R}_x(2\beta_\ell) \text{ on every qubit}} \underbrace{e^{-i\gamma_\ell \sum_i \hat{Z}_i\hat{Z}_{i+1}}}_{\hat{R}_{zz}(2\gamma_\ell) \text{ on every bond}} |+\rangle^{\otimes N},$$

with $n = 2L$ shared angles.

- Both exponents commute with the parity $\hat{\Pi} = \prod_i \hat{X}_i$, which flips every bit, and $\hat{\Pi}|+\rangle^{\otimes N} = |+\rangle^{\otimes N}$. Every HVA state is even, like the Ising ground state.
- Each layer has the form of a Trotter step of $e^{-i\hat{\mathcal{H}}t}$. Angles of a slow sweep from the field term to $\hat{\mathcal{H}}$ start training near the ground state (adiabatic theorem).

D. Wecker, M. B. Hastings and M. Troyer, PRA **92**, 042303 (2015), doi:10.1103/PhysRevA.92.042303
R. Wiersema *et al.*, PRX Quantum **1**, 020319 (2020), doi:10.1103/PRXQuantum.1.020319 H. F. Trotter, Proc. AMS **10**, 545 (1959), doi:10.1090/S0002-9939-1959-0108732-6 M. Born and V. Fock, Z. Phys. **51**, 165 (1928), doi:10.1007/BF01343193

The HVA Circuit



all $\hat{R}_{zz}$ of layer ℓ share the angle $2\gamma_\ell$, all $\hat{R}_x$ share $2\beta_\ell$, $n = 2L$

Four qubits, one layer shown. The tall boxes act on the even and odd bonds of the chain.

A Symmetry-Preserving Ansatz for the XXZ Chain

The XXZ chain

$$\hat{\mathcal{H}}_{\mathrm{XXZ}} = J \sum_i (\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \Delta \hat{Z}_i \hat{Z}_{i+1})$$

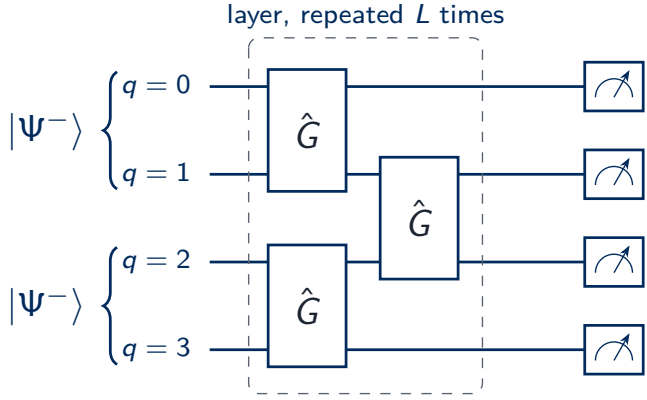
conserves the magnetisation $\sum_i \hat{Z}_i$, because $\hat{X}\hat{X} + \hat{Y}\hat{Y} = 2(|01\rangle\langle 10| + |10\rangle\langle 01|)$ only exchanges a 0 and a 1. At $\Delta = 1$ it is the Heisenberg antiferromagnet.

The SPA starts from singlets $|\Psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ on the pairs $(0,1), (2,3), \ldots$ and applies on even, then odd bonds

$$\hat{G} = e^{-i\theta_{xy}(\hat{X}\hat{X}+\hat{Y}\hat{Y})/2}\, e^{-i\theta_{zz}\hat{Z}\hat{Z}/2}.$$

$\hat{G}$ rotates $|01\rangle$ and $|10\rangle$ into each other and only adds phases to $|00\rangle$ and $|11\rangle$. No gate changes the number of ones, so every SPA state has $\sum_i \hat{Z}_i = 0$, the sector of the ground state. It has $n = 2(N-1)L$ angles.

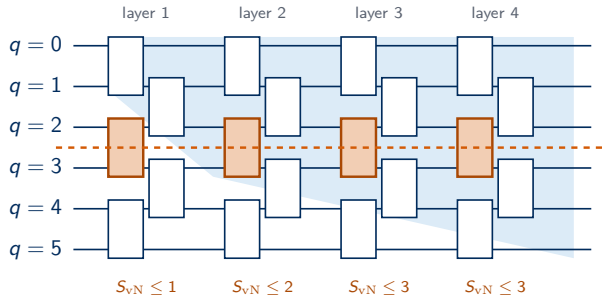
The SPA Circuit



each $\hat{G}$ has its own angles $(\theta_{xy}, \theta_{zz})$ and keeps the number of ones

Four qubits, one layer shown, starting from two singlets.

Entanglement Across the Middle Cut



shaded region, light cone of the first gate on $(0, 1)$

middle cut

one gate across the cut per layer (orange), at most 2 Schmidt terms each

$S_{\mathrm{vN}} \leq \min(L, N/2)$ bits

Gates inside one half leave the Schmidt coefficients across the cut unchanged. Each gate on the middle bond (orange) can add at most one bit, and the light cone of a gate grows by one bond per layer.

Entanglement Capacity of a Circuit

Only gates acting across the middle cut change the half-chain entropy

$$S_{\mathrm{vN}} = -\sum_k s_k^2 \log_2 s_k^2 \quad \text{(in bits)},$$

where s_k are the Schmidt coefficients across the cut. A gate written as $\hat{U} = \sum_{k=1}^{r} \hat{A}_k \otimes \hat{B}_k$ (left half, right half) multiplies the number of Schmidt terms by at most r. For $\widehat{\mathrm{CZ}}$ and $\hat{R}_{zz}$ we have $r = 2$,

$$\widehat{\mathrm{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z}, \qquad \hat{R}_{zz}(\theta) = \cos\tfrac{\theta}{2}\,\hat{I}\otimes\hat{I} - i\sin\tfrac{\theta}{2}\,\hat{Z}\otimes\hat{Z}.$$

One layer of the HEA or HVA contains one such gate on the middle bond, so after L layers at most 2^L Schmidt terms exist and

$$S_{\mathrm{vN}} \le \min(L,\, N/2) \text{ bits}.$$

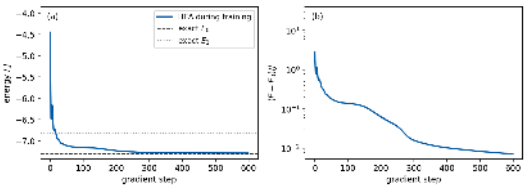
The six-qubit ground states studied below carry about one bit, which the bound allows from $L = 1$. Whether a circuit reaches a state also depends on the range of its correlations, which grows by one bond per layer.

6

The Variational Quantum Eigensolver

JAGIELLONIAN UNIVERSITY
IN KRAKOW

A First VQE Run



our simulation Energy against the Adam step for the HEA with $L = 4$ on the critical Ising chain, $N = 6$. Dashed lines mark E_0 and E_1, the inset shows $E - E_0$ on a log scale.

Initial state $|0 \ldots 0\rangle$ with small random angles. **Target** the ground state of $\hat{\mathcal{H}}$ at $h = J$. **Operations** the HEA gates, $n = 60$. **Success** a small relative energy error and, in simulation only, a fidelity F close to one.

Exact gradients from `jax.grad` and 600 Adam steps take the energy from -4.45 below E_1 within 17 steps. The final relative error is 1×10^{-3} and $F = 0.998$, consistent with $E - E_0 \geq \Delta E(1 - F)$.

Three Test Problems

We compare the ansätze on six-qubit chains whose exact ground states follow from sparse diagonalisation.

model	E_0	ΔE	S_{vN} (bits)	ansätze
Ising, critical, $h = J$	-7.2962	0.4821	0.473	HEA, HVA
Ising, ordered, $h = 0.5J$	-5.5220	0.0235	0.964	HEA, HVA
Heisenberg, $\Delta = 1$	-9.9743	1.9663	1.026	HEA, SPA

All ansätze have $L = 6$ layers and 30 two-qubit gates, with $n = 84$ (HEA), 12 (HVA) and 60 (SPA) angles. Each is trained with exact gradients and Adam for 600 iterations from 4 restarts, and we keep the restart with the lowest energy, as a VQE user would. We judge the result by

$$\delta E = \frac{E - E_0}{|E_0|}, \qquad F = |\langle \psi_0 | \psi \rangle|^2, \qquad \delta S = |S_{\mathrm{vN}}(\psi) - S_{\mathrm{vN}}(\psi_0)|.$$

Energy Error and Entanglement Error

We write a trial state as the ground state with a small admixture of a normalised state $|\phi\rangle \perp |\psi_0\rangle$,

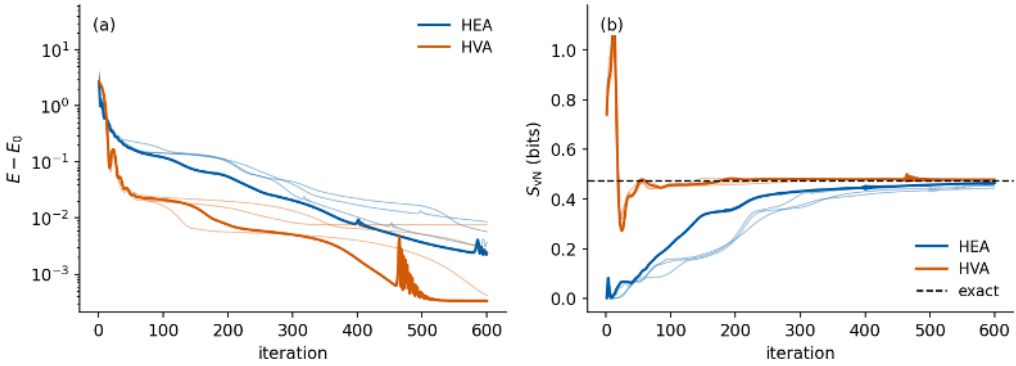
$$|\psi\rangle = \frac{|\psi_0\rangle + \epsilon|\phi\rangle}{\sqrt{1+\epsilon^2}}, \qquad 1 - F = \frac{\epsilon^2}{1+\epsilon^2}, \qquad E - E_0 = \frac{\epsilon^2\big(\langle\phi|\hat{\mathcal{H}}|\phi\rangle - E_0\big)}{1+\epsilon^2}.$$

The cross terms of the energy vanish, since $\hat{\mathcal{H}}|\psi_0\rangle = E_0|\psi_0\rangle$ and $\langle\phi|\psi_0\rangle = 0$. The energy error is therefore of second order in ϵ. The entropy has no stationary point at $|\psi_0\rangle$, and generically

$$S_{\mathrm{vN}}(\psi) - S_{\mathrm{vN}}(\psi_0) = \underbrace{\epsilon\, S_1}_{\text{first order}} + O(\epsilon^2), \qquad \delta S \propto \sqrt{1-F}, \qquad \delta E \propto 1 - F.$$

A state with a relative energy error of 10^{-4} can carry a visibly wrong entanglement, so we follow both quantities during training.

The Critical Ising Chain



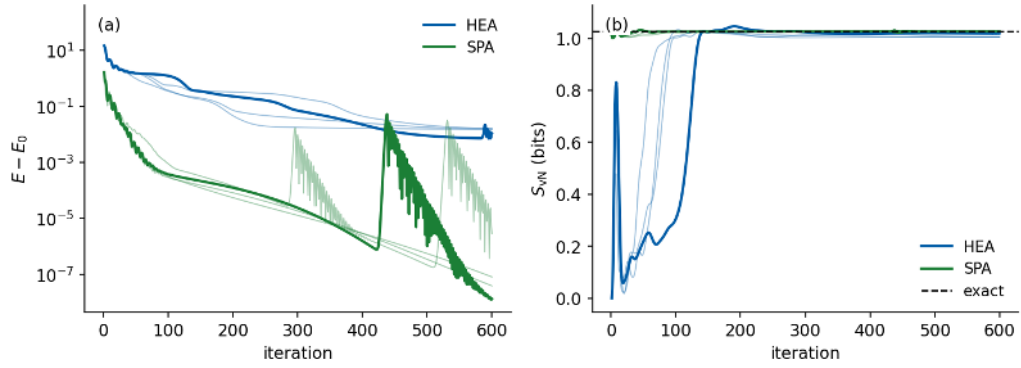
our simulation

(a) Energy error $E - E_0$ and (b) half-chain entropy against the iteration. Thick lines are the best restart, thin lines the other restarts, the dashed line the exact S_{vN}.

Energy and Entanglement at the Critical Point

- Both ansätze lower the energy error by three to four orders of magnitude.
- The HVA with 12 angles ends at $\delta E = 4.6 \times 10^{-5}$, below the HEA with 84 angles at 3.4×10^{-4}.
- The HEA builds its entanglement slowly from a product state. Its entropy error 1.1×10^{-2} bits exceeds the HVA value 4.4×10^{-3}, as the first-order scaling of the entropy error predicts.

The Heisenberg Chain



our simulation

(a) Energy error and (b) half-chain entropy against the iteration for the HEA and the SPA.

Singlets as a Starting Point

- The SPA reaches $\delta E = 1.3 \times 10^{-9}$ and keeps the exact entropy from the first iterations.
- It starts from singlets, which already carry one bit across the middle cut and lie in the sector $\sum_i \hat{Z}_i = 0$, so it only redistributes singlet correlations.
- The HEA has to build this structure from $|0 \ldots 0\rangle$ and ends at $\delta E = 7.3 \times 10^{-4}$. Late spikes in the curves are single large Adam steps.

The Ordered Ising Chain

At $h = 0.5J$ the ground state is close to a **cat state**, a superposition of two macroscopically different configurations,

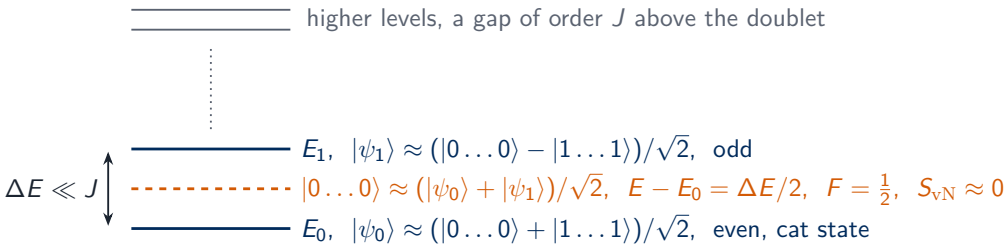
$$|\psi_0\rangle \approx \frac{|0\ldots0\rangle + |1\ldots1\rangle}{\sqrt{2}}, \qquad |\psi_1\rangle \approx \frac{|0\ldots0\rangle - |1\ldots1\rangle}{\sqrt{2}},$$

dressed by spin flips. The two levels form a doublet with the small gap $\Delta E = 0.0235J$. A trial state inside the doublet, $a|\psi_0\rangle + b|\psi_1\rangle$, has

$$\langle\hat{\Pi}\rangle = |a|^2 - |b|^2, \qquad E - E_0 = |b|^2 \Delta E = \underbrace{\frac{1 - \langle\hat{\Pi}\rangle}{2}}_{\text{odd weight}} \Delta E.$$

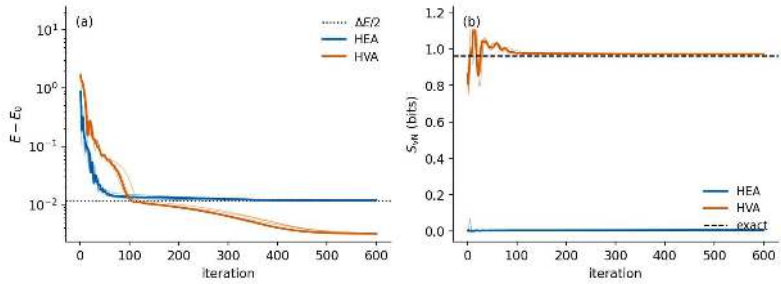
The symmetry-broken state $(|\psi_0\rangle + |\psi_1\rangle)/\sqrt{2} \approx |0\ldots0\rangle$ has $\langle\hat{\Pi}\rangle = 0$, $F = 1/2$, almost no entanglement, and an energy only $\Delta E/2$ above E_0. *Which of our two ansätze can prepare the symmetry-broken state?*

The Doublet of the Ordered Chain



The symmetry-broken state sits halfway between the two levels of the doublet. Its energy error $\Delta E/2$ is tiny on the scale of J, while its fidelity is one half and its entanglement is lost.

The Symmetry-Broken Trap

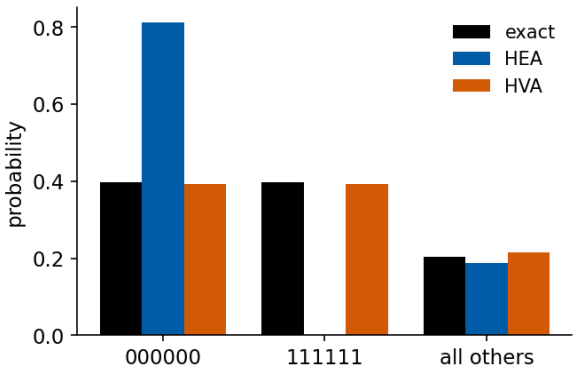


our simulation

(a) Energy error with the dotted line $\Delta E/2$ and (b) half-chain entropy against the iteration, ordered chain $h = 0.5J$.

Every HEA restart descends into the branch near $|0 \ldots 0\rangle$ where it starts, with $F = 0.5093$ and $S_{\mathrm{vN}} = 0.006$. Its energy error 0.0118 sits on $\Delta E/2 = 0.0117$. The parity-conserving HVA reaches the cat state, $F = 0.9988$.

Bit Strings of the Trapped State

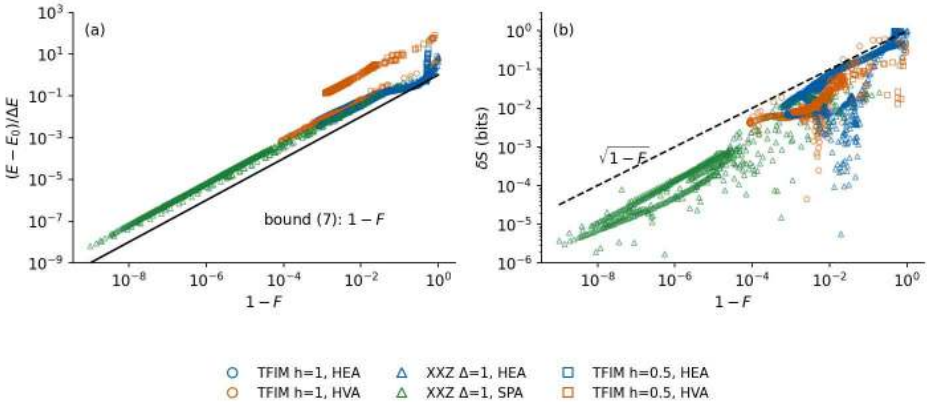


our simulation Probabilities of 000000 and 111111 in the trained HEA and HVA states and in the exact ground state.

A device measuring in the computational basis sees the trap directly. The exact and HVA states return 000000 and 111111 in nearly equal shares, the HEA state returns 000000 with probability 0.81 and 111111 almost never.

Measurements along z alone cannot tell the cat state from an equal mixture of the branches. The parity $\langle \hat{\Pi} \rangle$, measured in the all-$x$ setting, decides. The HEA state has $\langle \hat{\Pi} \rangle = +0.019$.

Energy Error Against State Error



our simulation

Every iteration of the six runs, 14400 states. (a) Energy error in units of the gap with the bound $(E - E_0)/\Delta E = 1 - F$ (solid). (b) Entropy error with the line $\sqrt{1 - F}$ (dashed).

Two Error Scalings

- All points obey the variational bound and follow a slope of one, while the entropy errors follow the slope one half.
- At small infidelity the entropy is therefore the more sensitive test of the state.
- The symmetry-broken HEA states touch the bound near $1 - F = 1/2$, where nearly all the error sits in the first excited state.

Comparison of the Ansätze

model	ansatz	n	n_{rot}	circuit runs per gradient	δE	F	δS (bits)
Ising $h = J$	HEA	84	84	336	3.4×10^{-4}	0.9994	1.1×10^{-2}
	HVA	12	66	264	4.6×10^{-5}	0.9999	4.4×10^{-3}
Ising $h = 0.5J$	HEA	84	84	336	2.1×10^{-3}	0.5093	9.6×10^{-1}
	HVA	12	66	264	5.7×10^{-4}	0.9988	8.0×10^{-3}
Heisenberg	HEA	84	84	504	7.3×10^{-4}	0.9992	7.6×10^{-3}
	SPA	60	90	540	1.3×10^{-9}	1.0000	4.1×10^{-6}

n_{rot} counts parametrised Pauli rotations, and one hardware gradient needs $2n_{\mathrm{rot}}$ shifted circuits in G settings, $2n_{\mathrm{rot}}G$ runs of M shots ($G = 2$ for Ising, 3 for Heisenberg). Best restart, $L = 6$, 30 two-qubit gates.

In our runs (four restarts, 600 Adam steps), at equal depth and two-qubit gate count, the ansätze built for their Hamiltonian needed fewer angles and reached better states. Their advantage is tied to one model and one symmetry sector.

Excited States by Variational Quantum Deflation

After VQE has produced $|\tilde{\psi}_0\rangle$, a second run minimises

$$C(\boldsymbol{\theta}) = E(\boldsymbol{\theta}) + \underbrace{\lambda\,|\langle\tilde{\psi}_0|\psi(\boldsymbol{\theta})\rangle|^2}_{\text{penalty on ground-state overlap}} .$$

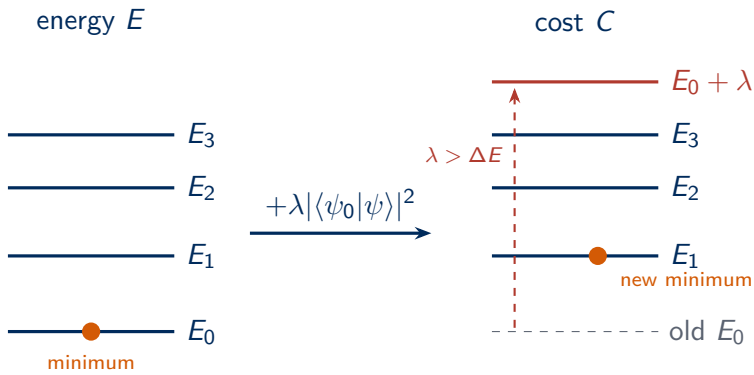
For an exact $|\tilde{\psi}_0\rangle$ a trial state $\sum_n c_n|n\rangle$ has the cost $\sum_n |c_n|^2 E_n + \lambda|c_0|^2$, the energy of a Hamiltonian with the ground level moved to $E_0 + \lambda$. For $\lambda > \Delta E$ its minimum is E_1, reached at the first excited state.

On a device the overlap is the probability of $0\ldots0$ after $\hat{U}(\boldsymbol{\theta})$ followed by $\hat{U}(\tilde{\boldsymbol{\theta}}_0)^\dagger$, since $|\langle 0\ldots0|\hat{U}(\tilde{\boldsymbol{\theta}}_0)^\dagger \hat{U}(\boldsymbol{\theta})|0\ldots0\rangle|^2 = |\langle\tilde{\psi}_0|\psi(\boldsymbol{\theta})\rangle|^2$.

Errors of $|\tilde{\psi}_0\rangle$ propagate into the excited state, because the penalty acts on the approximate ground state.

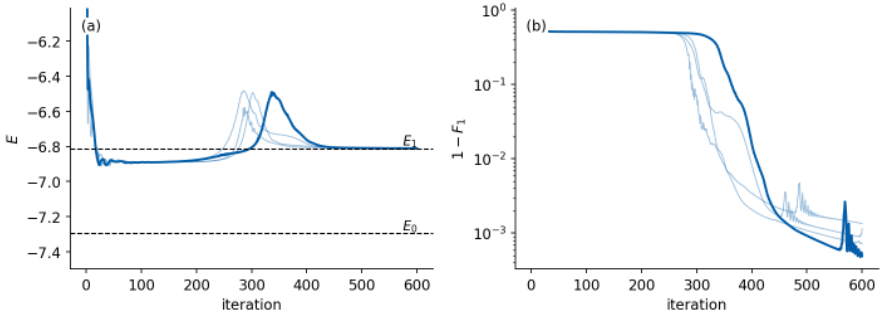
O. Higgott, D. Wang and S. Brierley, Quantum **3**, 156 (2019), doi:10.22331/q-2019-07-01-156

Lifting the Ground Level



The penalty acts only on the overlap with the ground state, so it raises E_0 to $E_0 + \lambda$ and leaves the other levels in place. For $\lambda > \Delta E$ the lowest cost is E_1.

The First Excited State of the Critical Chain



our simulation

(a) Energy against the iteration with the exact E_0, E_1 dashed. (b) Infidelity $1 - F_1$ with the exact first excited state.

With the HEA and $\lambda = 2J$ the run reaches $F_1 = 0.9995$, and the gap 0.4818 agrees with the exact 0.4821. The HVA, confined to even states, cannot reach the odd $|\psi_1\rangle$.

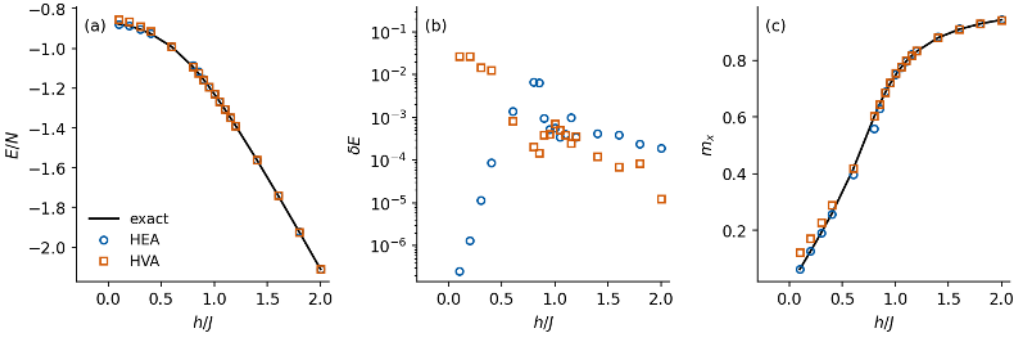
Following the Ground State Across the Transition

We train the HEA and the HVA on the Ising chain with $N = 8$ at 18 fields between $h = 0.1J$ and $2J$, every field independently, with $L = 8$, 56 two-qubit gates, $n = 144$ (HEA) and 16 (HVA), 2 restarts and 800 Adam iterations. Exact references come from Lanczos with a matrix-free $\hat{\mathcal{H}}|\psi\rangle$.
Besides E/N, δE, S_{vN} and F we record two magnetisations,

$$m_x = \frac{1}{N}\sum_i \langle \hat{X}_i \rangle, \qquad m_z = \sqrt{\Big\langle \Big(\frac{1}{N}\sum_i \hat{Z}_i\Big)^2 \Big\rangle}.$$

- m_x is the alignment with the field and tends to one for $h \gg J$.
- $\hat{\Pi}\hat{Z}_i\hat{\Pi} = -\hat{Z}_i$, so $\langle \hat{Z}_i \rangle = 0$ in every parity eigenstate. The parity-invariant m_z tends to one in the ordered phase and to about $1/\sqrt{N}$ in the paramagnet.
- m_z is nearly equal in the cat state and in either of its branches, so it cannot detect the trap. The entropy and the fidelity can.

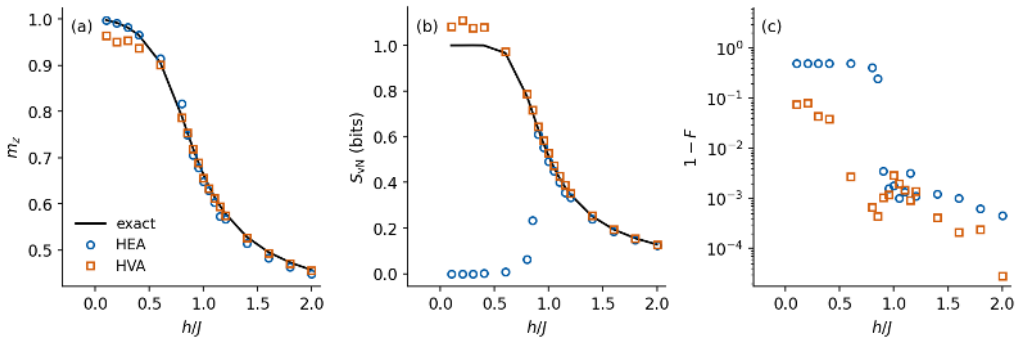
The Ising Phase Diagram at $N = 8$, Energy and Field



our simulation

Exact state-vector simulations against h/J, (a) energy per site, (b) relative energy error δE, (c) transverse magnetisation m_x. Black lines are exact diagonalisation, markers the best restart of each ansatz.

The Ising Phase Diagram at $N = 8$, Order and Entanglement



our simulation

Exact state-vector simulations against h/J, (a) order parameter m_z, (b) half-chain entropy S_{vN}, (c) infidelity $1 - F$.

Failure Modes Across the Transition

- The energies and m_x of both ansätze follow the exact curves on the scale of the figures, and so does m_z of the HEA.
- Deep in the ordered phase the HEA has the *smallest* energy errors of the scan, $\delta E = 1.3 \times 10^{-6}$ at $h = 0.2J$, with $F = 0.500$ and $S_{\mathrm{vN}} = 0.000$. It sits in a symmetry-broken branch whose error shrinks with the doublet splitting. Its fidelity is near $1/2$ up to $h = 0.6J$, stays below 0.99 up to $h = 0.85J$ and exceeds 0.99 from $h = 0.9J$.
- The HVA keeps even parity, follows the exact entropy through the transition and reaches $F > 0.99$ from $h = 0.6J$. For $h \leq 0.4J$ its fidelity is 0.92 to 0.96 and its energy error exceeds that of the trapped HEA. The same circuit exceeds $F = 0.99$ from $h = 0.6J$, so the deficit is confined to the deep ordered phase.

Energy hides the state

Ranked by energy the HEA wins in the ordered phase, ranked by fidelity or entanglement the HVA wins. Near the critical point both ansätze must build the entanglement of the critical ground state, and the entropy errors grow there for the HEA.

The XXZ Chain in a Transverse Field

We next scan a two-dimensional parameter space, with $N = 8$,

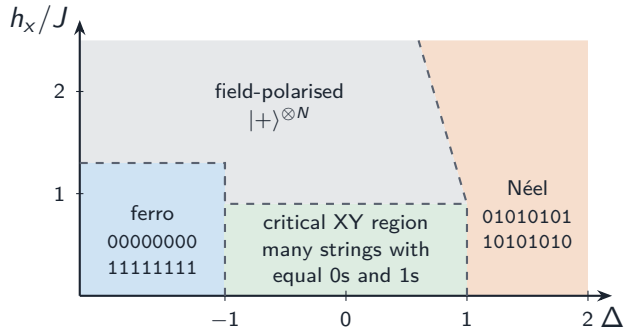
$$\hat{\mathcal{H}} = J \sum_{i=0}^{N-2} \left(\hat{X}_i \hat{X}_{i+1} + \hat{Y}_i \hat{Y}_{i+1} + \Delta \hat{Z}_i \hat{Z}_{i+1} \right) - h_x \sum_{i=0}^{N-1} \hat{X}_i .$$

The field flips single bits and breaks the conservation of $\sum_i \hat{Z}_i$. Only the parity $\hat{\Pi}$ remains, so the ground state builds its entanglement from superpositions across magnetisation sectors.

Order along z is measured by the parity-invariant magnetisations

$$m_{\mathrm{stag}} = \sqrt{\left\langle \left(\frac{1}{N} \sum_i (-1)^i \hat{Z}_i \right)^2 \right\rangle} \ \text{(Néel = 1)}, \qquad m_z = \sqrt{\left\langle \left(\frac{1}{N} \sum_i \hat{Z}_i \right)^2 \right\rangle} \ \text{(ferro = 1)}.$$

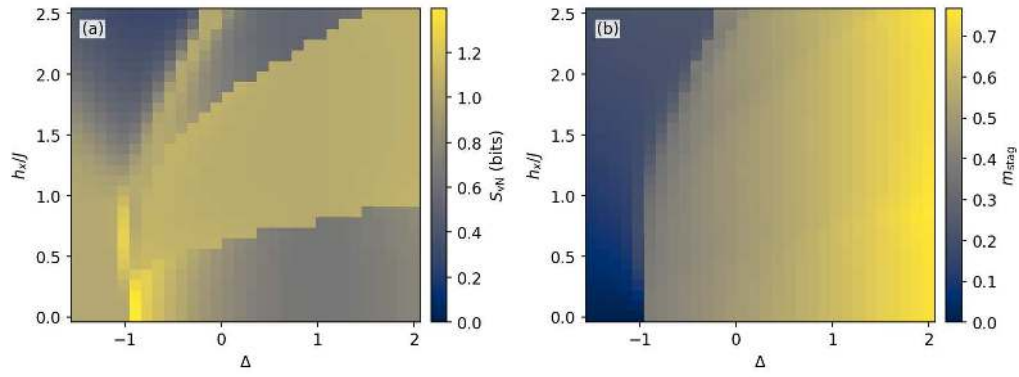
Regions of the XXZ Chain in a Field



schematic, on eight sites the regions are smooth crossovers

The ferromagnet ($\Delta < -1$) is a cat state of 00000000 and 11111111, the Néel state ($\Delta > 1$) favours 01010101 and 10101010, and the XY region between them is critical. Large h_x turns the spins towards $+x$. At $h_x = 2.5J$, m_x reaches 0.85 for $\Delta < 0$ and stays ≤ 0.5 for $\Delta > 0$.

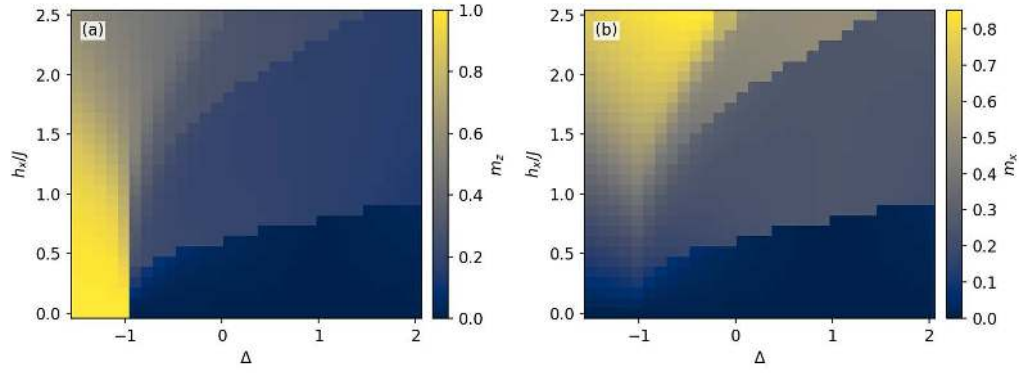
Exact Maps, Entanglement and Staggered Order



our simulation

Exact ground state (matrix-free Lanczos, exact state vectors) on a 30×30 grid of (Δ, h_x). (a) Half-chain entropy S_{vN} in bits, (b) staggered magnetisation m_{stag}.

Exact Maps, Magnetisations



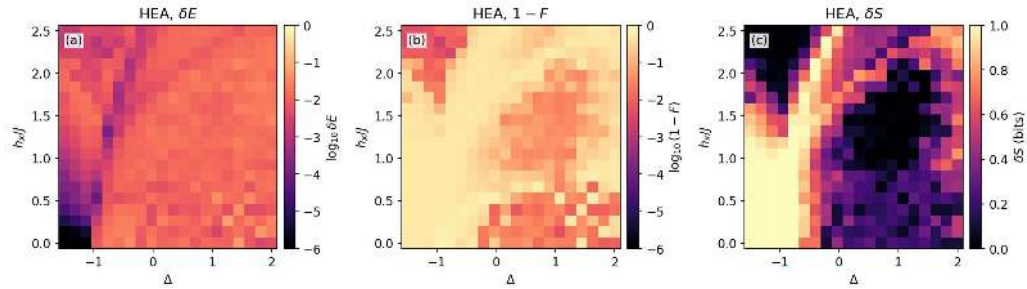
our simulation

Same grid. (a) Ferromagnetic order parameter m_z, (b) transverse magnetisation m_x.

Features of the Exact Maps

- The ferromagnet in the lower left carries one bit, as the cat state does, with $m_z \approx 1$.
- The staggered order grows with Δ, reduced below one by quantum fluctuations of eight spins.
- The XY region has $S_{\mathrm{vN}} = 0.841$ bits.
- On the antiferromagnetic side the field changes the ground state in steps. These are finite-size level crossings between the two parity sectors and move with N.

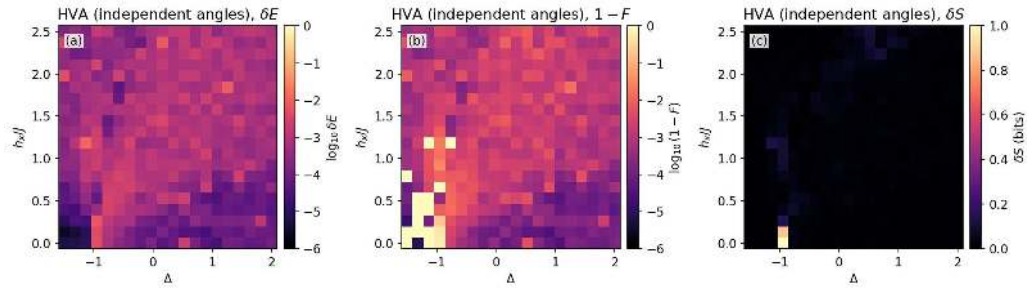
Error Maps of the HEA



our simulation

Exact state-vector simulations. (a) $\log_{10}\delta E$, (b) $\log_{10}(1-F)$, (c) δS in bits, dark is small. $L = 6$, 42 two-qubit gates, $n = 112$, 20×20 points, two starts, 300 iterations.

Error Maps of the HVA with Independent Angles



our simulation

Same quantities for the SPA gates of the XXZ chain followed by $\hat{R}_x$ on every qubit, each gate with its own angle, $n = 132$, 42 two-qubit gates, same budget.

Failures on the Grid

- The HVA with independent angles, trained once from the singlet product (even parity) and once from the same product with the first pair replaced by the triplet, $\hat{Z}_0$ applied to it (odd parity), reaches $F > 0.99$ almost everywhere. It fails in the ferromagnetic corner and along $\Delta \approx -1$, where the two parity sectors are nearly degenerate and the lower trained energy can belong to the wrong sector. Near $\Delta = -1$ at small h_x it also fails within the correct sector, with large entropy errors.
- The HEA reaches $F > 0.99$ at only a few points. Deep in the ferromagnet it has the smallest energy errors of the map with $F = 1/2$ and an entropy error of one bit, the symmetry-broken branch again. Elsewhere its energy error stays near 10^{-2} and its entropy lies below the exact value, so it builds too little entanglement within this budget.
- At 34 of the 400 points the HEA has $\delta E < 10^{-3}$ together with $F < 0.6$.

Consequence

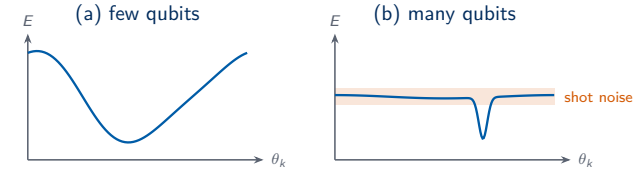
A small energy error does not certify the trained state. The fidelity and the entanglement entropy reveal the failures.

7

Barren Plateaus

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Gradients at Random Angles



Training often starts from angles drawn uniformly from $[-\pi, \pi]$. Along one angle every cost is $a + b\cos(\theta_k - c)$, so its derivative averages to zero, and the typical size of a gradient component is set by its variance over random angles,

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \big\langle (\partial_k C)^2 \big\rangle_{\boldsymbol{\theta}}.$$

For deep random circuits this variance decays exponentially with N. The landscape is flat almost everywhere, a **barren plateau**, and a device must resolve exponentially small gradients above the shot noise.

J. R. McClean *et al.*, Nat. Commun. **9**, 4812 (2018), doi:10.1038/s41467-018-07090-4

Concentration in Random States

Deep circuits of random gates reproduce, for first and second moments, a unitary drawn uniformly from all unitaries on $d = 2^N$ dimensions (Haar measure), a *unitary 2-design*. In a random state every expectation value concentrates around its mean,

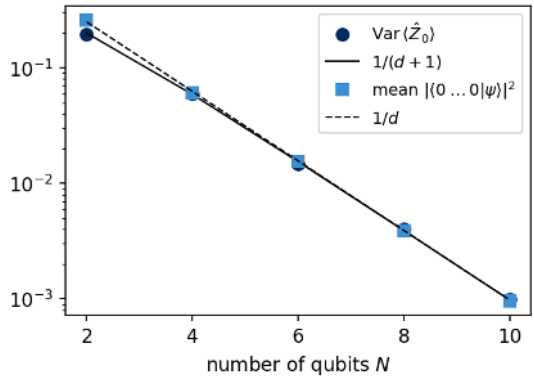
$$\mathrm{Var}\,\langle \hat{P} \rangle = \frac{1}{d+1} \approx 2^{-N} \quad (\hat{P} \neq \hat{I}), \qquad \langle |\langle 0 \ldots 0 | \psi \rangle|^2 \rangle = \frac{1}{d},$$

since $\langle \hat{Z}_0 \rangle$ is a difference of two sums of $d/2$ random weights. If all gates after the differentiated rotation act as a Haar-random unitary, a cost $C = \langle \hat{O} \rangle$ has

$$\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k C] = \frac{\mathrm{Tr}(\hat{O}_0^2)}{2(d^2-1)}, \qquad \hat{O}_0 = \hat{O} - \frac{\mathrm{Tr}\,\hat{O}}{d}\,\hat{I}.$$

- The global cost $\hat{O} = \hat{I} - |0 \ldots 0\rangle\langle 0 \ldots 0|$ gives $\mathrm{Var} \approx 2^{-2N-1}$.
- The Ising energy, a sum of Pauli strings, gives $\mathrm{Var} \approx [(N-1)J^2 + Nh^2]/2^{N+1}$.

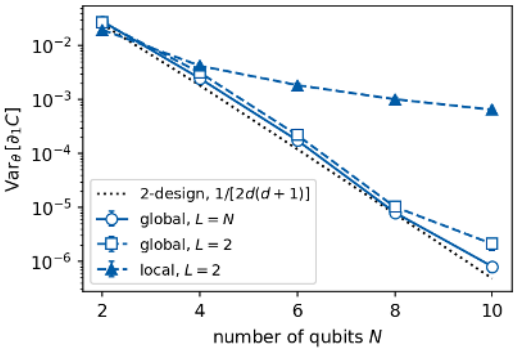
Random States Concentrate



our simulation

Variance of $\langle \hat{Z}_0 \rangle$ and mean of $|\langle 0 \ldots 0 | \psi \rangle|^2$ over 2000 random states against N, with the lines $1/(d+1)$ and $1/d$, $d = 2^N$. Both halve with every added qubit.

Global and Local Costs



our simulation Variance of $\partial C/\partial\theta_1$ over 300 random angle vectors for the HEA, against N, with the 2-design value of the global cost.

M. Cerezo *et al.*, Nat. Commun. **12**, 1791 (2021), doi:10.1038/s41467-021-21728-w

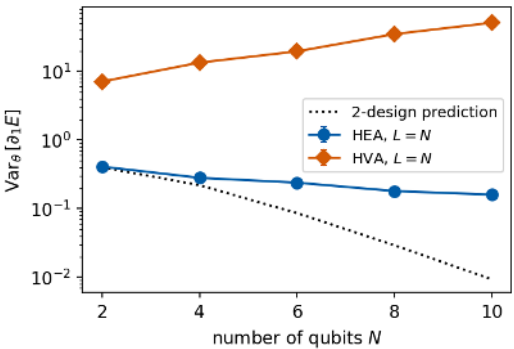
Both costs vanish only in $|0\ldots0\rangle$,

$$C_G = 1 - |\langle 0\ldots0|\psi\rangle|^2,$$

$$C_L = \frac{1}{N}\sum_q \frac{1 - \langle \hat{Z}_q\rangle}{2}.$$

The global cost decays as 2^{-cN} with $c = 1.96$ at $L = N$, the 2-design value, and $c = 1.81$ already at $L = 2$. The local cost at $L = 2$ sees only the light cone of the differentiated gate, and $N^2\,\mathrm{Var}$ stays constant. At $N = 10$ its variance is 311 times larger.

The Energy Gradient of HEA and HVA



our simulation Variance of the Ising energy gradient at $h = J$ for the HEA and the HVA at depth $L = N$, with the 2-design value.

M. Ragone *et al.*, Nat. Commun. **15**, 7172 (2024), doi:10.1038/s41467-024-49909-3 M. Larocca *et al.*, Nat. Rev. Phys. **7**, 174 (2025), doi:10.1038/s42254-025-00813-9

The energy is a sum of terms on one or two neighbouring qubits. For the HEA at depth N its gradient decays far more slowly than the 2-design curve at these sizes, because the circuit has only just connected the two ends of the chain.

The HVA variance grows with N. One shared angle changes all $2N - 1$ terms together, and its gates generate a family of unitaries whose dimension grows only polynomially with N, the dimension of its dynamical Lie algebra.

Shots Needed to Resolve a Gradient

A parameter-shift component estimated with M shots per setting has the statistical variance $\sigma^2/(2M)$, with the single-shot energy variance σ^2. The typical gradient is resolved when this equals $\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E]$,

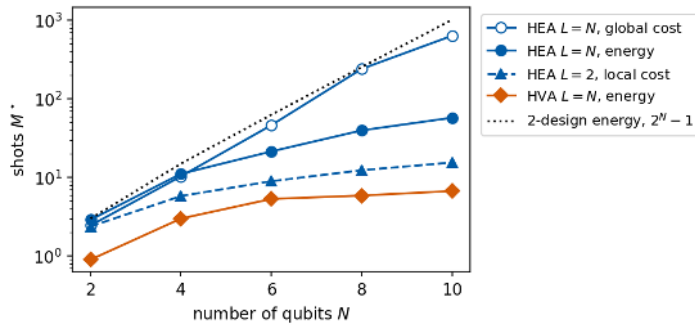
$$M^\star = \frac{\sigma^2}{2\,\mathrm{Var}_{\boldsymbol{\theta}}[\partial_k E]} \xrightarrow{\text{2-design}} d - 1 = 2^N - 1.$$

	HEA, $N = 10$	2-design, $N = 10$	2-design, $N = 20$	2-design, $N = 50$
$M^\star$	58	1023	1×10^6	1×10^{15}

For one component of one gradient, a 2-design doubles the shot count with every qubit. An HVA angle enters $N - 1$ gates, and each of its $2(N - 1)$ shifted circuits needs only about 6.7 shots at $N = 10$. A better optimiser cannot remove a plateau, since every optimiser on a device must resolve energy differences above the shot noise.

our simulation

Shots Against System Size



our simulation

Shots $M^\star$ per setting and shifted circuit needed to resolve a typical gradient component, from the measured variances, with the 2-design line $2^N - 1$.

The global cost follows the exponential 2-design line, while the local cost and the HVA need only a few shots at these sizes.

Noise-Induced Plateaus and Remedies

Depolarising noise after each layer replaces part of the state by the maximally mixed state. The landscape contracts as $(1-\varepsilon)^{\mathrm{depth}}$ for an error probability ε per layer, and for depth growing with N the gradient vanishes exponentially, also for structured ansätze.
A circuit that can prepare almost any state has small gradients, so expressibility and trainability have to be balanced. Known remedies restrict the states that the circuit explores.

- Local costs at logarithmic depth keep polynomially small gradients, and shallow circuits also limit the noise.
- Structured ansätze that respect symmetries, such as the HVA, stay in a small part of the state space.
- Initial angles that make the circuit the identity, and layerwise training, start away from the random region.

S. Wang *et al.*, Nat. Commun. **12**, 6961 (2021), doi:10.1038/s41467-021-27045-6 E. Grant *et al.*, Quantum **3**, 214 (2019), doi:10.22331/q-2019-12-09-214 A. Skolik *et al.*, Quantum Mach. Intell. **3**, 5 (2021), doi:10.1007/s42484-020-00036-4 Z. Holmes *et al.*, PRX Quantum **3**, 010313 (2022), doi:10.1103/PRXQuantum.3.010313

8

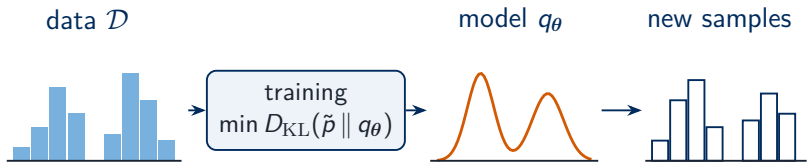
Circuits as Probability Models



JAGIELLONIAN UNIVERSITY
IN KRAKÓW

Generative Modelling

A data set $\mathcal{D}$ contains independent samples of an unknown distribution p. The goal is a model q_θ that can be sampled and comes close to p, so that its new samples look like the data without copying them.



classical models
histogram, Boltzmann machine, autoregressive network, GAN, VAE

Born machine
$q_\theta(\mathbf{b}) = |\langle \mathbf{b}|\hat{V}(\boldsymbol{\theta})|0\ldots0\rangle|^2$
one measurement = one sample

A register measured in the computational basis returns Born-rule samples, so a parametrised circuit defines such a model.

Encoding a Continuous Variable

A register returns bit strings, so the data must be bit strings. Discrete data such as spin configurations are bit strings already. A continuous $x \in [x_{\min}, x_{\max}]$ is cut into 2^N bins of equal width w,

$$w = \frac{x_{\max} - x_{\min}}{2^N}, \qquad k(x) = \left\lfloor \frac{x - x_{\min}}{w} \right\rfloor = \underbrace{\sum_{q=0}^{N-1} b_q \, 2^{N-1-q}}_{\text{binary, qubit 0 most significant}}, \qquad x_k = x_{\min} + \left(k + \tfrac{1}{2}\right) w.$$

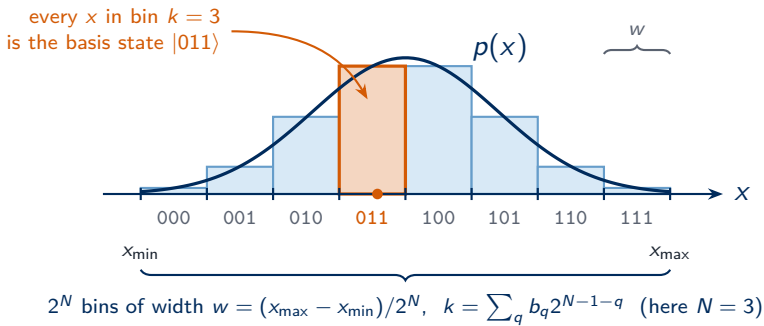
Every x in bin k corresponds to the basis state $|\mathbf{b}\rangle$, and a measured bit string is read back as the bin centre x_k.

Example, $N = 3$ on $[-4, 4]$

$w = 1$, eight bins $000, \ldots, 111$. The value $x = -0.37$ has $k = \lfloor 3.63 \rfloor = 3 = 0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1$, the bit string 011 and the state $|011\rangle$. A measured 011 is read as $x_3 = -0.5$.

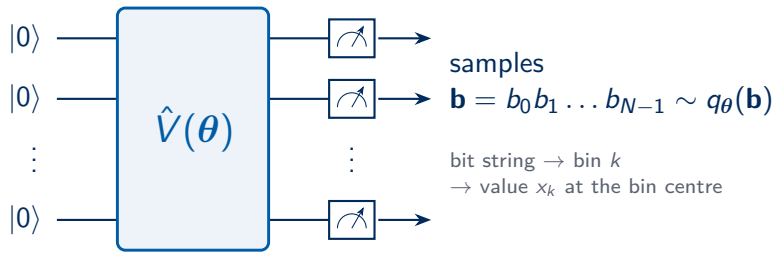
The discretisation loses the position inside a bin, at most $w/2$. Each added qubit halves w. The data define the empirical distribution $\tilde{p}(\mathbf{b}) = n(\mathbf{b})/|\mathcal{D}|$, with $n(\mathbf{b})$ the count of $\mathbf{b}$ in the data.

Binning a Continuous Variable



Bin $k = 3 = 0 \cdot 4 + 1 \cdot 2 + 1 \cdot 1$ is the bit string 011 and the basis state $|011\rangle$. Every value of x in that bin is encoded by the same state, and a measured 011 is read back as the bin centre.

Sampling from the Register



The circuit $\hat{V}(\boldsymbol{\theta})$ prepares a state on $|0\rangle^{\otimes N}$. Each measurement of all qubits returns one bit string, one sample of the model, which is read back as a value of x.

The Born Machine

A circuit $\hat{V}(\boldsymbol{\theta})$ on $|0\rangle^{\otimes N}$ followed by a measurement of every qubit returns $\mathbf{b}$ with probability

$$q_{\boldsymbol{\theta}}(\mathbf{b}) = \left|\langle \mathbf{b}|\hat{V}(\boldsymbol{\theta})|0\rangle^{\otimes N}\right|^2, \qquad \sum_{\mathbf{b}} q_{\boldsymbol{\theta}}(\mathbf{b}) = 1 \text{ for every } \boldsymbol{\theta}.$$

A model whose probabilities are squared amplitudes is a **Born machine**. Matrix product states first served as the source of amplitudes, a parametrised circuit gives the quantum circuit Born machine.

- Sampling is native. One run and one measurement draw one bit string and, through x_k, one value of x.
- The probabilities are known exactly in simulation. A device estimates them only as frequencies $\tilde{q}(\mathbf{b}) = n(\mathbf{b})/M$ over M shots.

Z.-Y. Han *et al.*, PRX **8**, 031012 (2018), doi:10.1103/PhysRevX.8.031012 S. Cheng, J. Chen and L. Wang, Entropy **20**, 583 (2018), doi:10.3390/e20080583 J.-G. Liu and L. Wang, PRA **98**, 062324 (2018), doi:10.1103/PhysRevA.98.062324 M. Benedetti *et al.*, npj Quantum Inf. **5**, 45 (2019), doi:10.1038/s41534-019-0157-8 B. Coyle *et al.*, npj Quantum Inf. **6**, 60 (2020), doi:10.1038/s41534-020-00288-9

The Loss Function

The Kullback–Leibler divergence (in nats) is non-negative and vanishes only for $q_\theta = p$,

$$D_{\mathrm{KL}}(p \,\|\, q_\theta) = \sum_{\mathbf{b}} p(\mathbf{b}) \ln \frac{p(\mathbf{b})}{q_\theta(\mathbf{b})}.$$

With the empirical $\tilde{p}$ in place of p, the logarithm splits, and the sum over bit strings weighted by $n(\mathbf{b})/|\mathcal{D}|$ becomes a sum over the data,

$$D_{\mathrm{KL}}(\tilde{p} \,\|\, q_\theta) = \underbrace{\sum_{\mathbf{b}} \tilde{p}(\mathbf{b}) \ln \tilde{p}(\mathbf{b})}_{-H(\tilde{p}),\ \text{independent of } \theta} \quad \underbrace{-\frac{1}{|\mathcal{D}|} \sum_{i=1}^{|\mathcal{D}|} \ln q_\theta(\mathbf{b}^{(i)})}_{\mathcal{L}_{\mathrm{NLL}}(\theta),\ \text{negative log-likelihood}} .$$

Minimising the KL divergence to the data and maximising the likelihood of the data are the same task with the same gradient.

S. Kullback and R. A. Leibler, Ann. Math. Statist. **22**, 79 (1951), doi:10.1214/aoms/1177729694

Training a Born Machine

In VQE the bit strings serve to estimate an energy and the state is the goal. In a Born machine the distribution of the bit strings is the goal and the samples are the output.

- The loss diverges when $q_{\boldsymbol{\theta}} \to 0$ on a data bit string, so training starts near the uniform distribution (each qubit near $|+\rangle$).
- It needs $q_{\boldsymbol{\theta}}(\mathbf{b})$ itself, exact in simulation but only a shot frequency on a device, where an empty bin makes $\ln \tilde{q}$ infinite.
- Gradients come from `jax.grad` in simulation and from the parameter-shift rule on a device, since $q_{\boldsymbol{\theta}}(\mathbf{b}) = \langle \mathbf{b} | \rho | \mathbf{b} \rangle$ is an expectation value.
- Our circuits interleave rotation blocks $\bigotimes_q \hat{R}_z(\phi_{\ell q}) \hat{R}_y(\vartheta_{\ell q})$ with $\widehat{\mathrm{CZ}}$ chains, L entangling layers and $n = 2N(L+1)$ angles, trained with Adam.
- We judge a model by $D_{\mathrm{KL}}(p \,\|\, q_{\boldsymbol{\theta}})$ to the exact target, which tests generalisation beyond the data, and by the total variation distance $d_{\mathrm{TV}}(p, q) = \frac{1}{2} \sum_{\mathbf{b}} |p(\mathbf{b}) - q(\mathbf{b})|$, which stays finite on empty bins.

Product and Entangling Circuits

Without entangling gates every qubit ends in its own state $\cos\frac{\vartheta_q}{2}|0\rangle + e^{i\phi_q}\sin\frac{\vartheta_q}{2}|1\rangle$, and the Born probability factorises,

$$q_\theta(\mathbf{b}) = \prod_{q=0}^{N-1} q_q(b_q), \qquad q_q(0) = \cos^2\tfrac{\vartheta_q}{2}, \quad q_q(1) = \sin^2\tfrac{\vartheta_q}{2}.$$

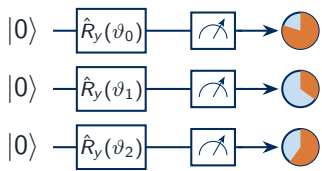
Every bit is an independent biased coin. The logarithm of a product is a sum, and the KL divergence is smallest for the product of the marginals $\tilde{p}_q(b_q)$ of the data distribution $\tilde{p}$,

$$D_{\mathrm{KL}}\Big(\tilde{p}\,\Big\|\,\prod_q q_q\Big) \;\geq\; \underbrace{-H(\tilde{p}) - \sum_q \sum_{b_q} \tilde{p}_q(b_q)\ln\tilde{p}_q(b_q)}_{\text{product bound, no training goes below}}.$$

- Entanglers correlate the bits. Deep fully trainable circuits meet barren plateaus, and one remedy fixes the entanglers and trains only one-qubit rotations.

Independent Coins and Correlated Bits

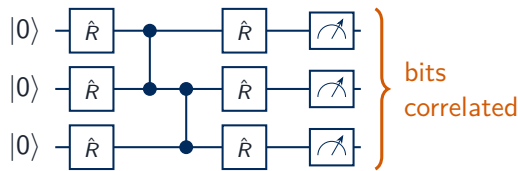
(a) product circuit



independent biased coins

$q(\mathbf{b}) = \prod_q q_q(b_q)$, no correlations

(b) circuit with entanglers

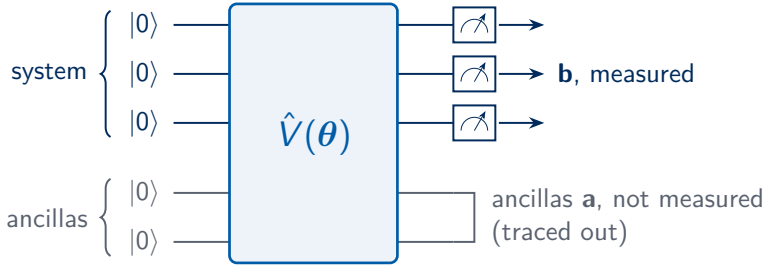


$\widehat{\mathrm{CZ}}$ gates create the correlations that a two-peak target needs

best product model = product of the data marginals $p_q(b_q)$

D_{KL} never falls below the product bound

Ancillas and Mixtures

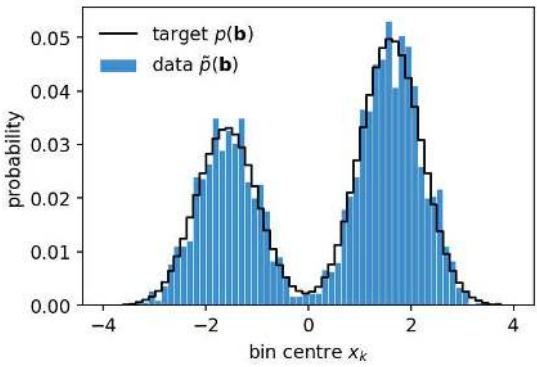


$$q(\mathbf{b}) = \sum_{\mathbf{a}} |\langle \mathbf{b}, \mathbf{a} | \psi(\boldsymbol{\theta}) \rangle|^2 = \langle \mathbf{b} | \hat{\rho}_{\text{sys}} | \mathbf{b} \rangle$$

a mixture of Born distributions, one for each ancilla string **a**

A pure state of the measured qubits limits the model. Ancilla qubits take part in the circuit and are not measured, so the Born probabilities are summed over their bits and the model becomes a mixture of Born distributions.

A Two-Peak Target



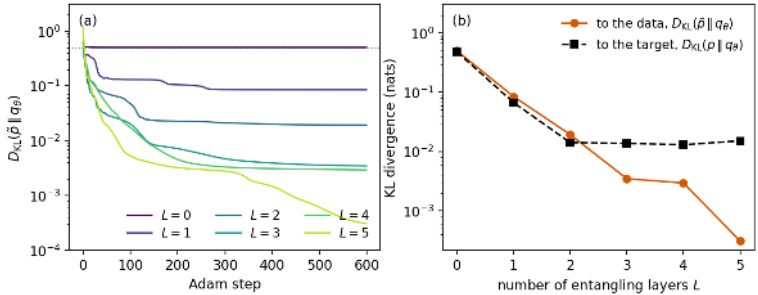
our simulation Bars, empirical distribution of the data. Line, exact bin probabilities of the target.

The target is a mixture of two Gaussians with weights 0.4 and 0.6, centres ∓ 1.6 and width 0.6 on $[-4, 4]$. On $N = 6$ qubits the 64 bins have $w = 0.125$.

The learner sees $|\mathcal{D}| = 3000$ values. Their total variation distance to the target, $d_{\mathrm{TV}}(p, \tilde{p}) = 0.065$, sets the statistical error of the data.

The left peak sits near bin 19, 010011, the right near bin 44, 101100. The leading bits of the two peaks are opposite, so the bits are strongly correlated.

Training Against Depth

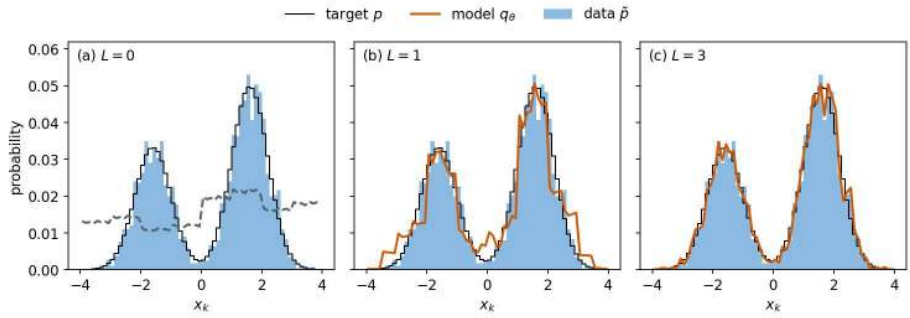


our simulation

(a) KL divergence to the data against the Adam step, best of three starts at each depth L, product bound dotted.
(b) Final KL divergence to the data (circles) and to the exact target (squares).

The product circuit stops at the bound 0.500, and one entangling layer brings the divergence to the target down to 0.0669. Beyond $L = 2$ the loss on the data keeps falling while the divergence to the target saturates near 0.014, since deeper circuits fit the fluctuations of the finite data set (overfitting). With ten times more data it drops to about 0.0019 at $L = 3$.

Trained Distributions

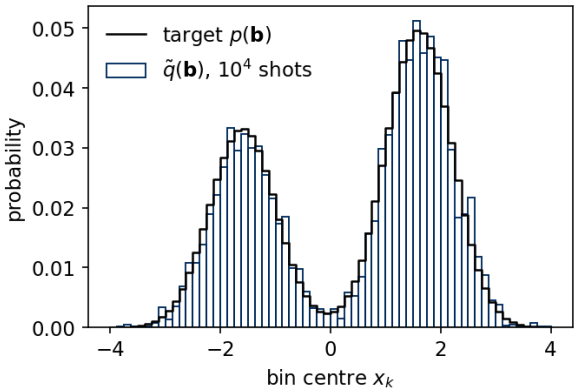


our simulation

Trained model (line) and target for (a) the product circuit, (b) $L = 1$, (c) $L = 3$.

The product model spreads probability over the interval and misses both peaks. One layer places the peaks with roughly the right weights, three layers follow the target in peaks and tails.

Samples from the Trained Circuit

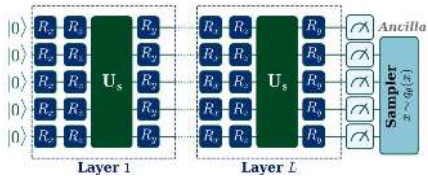


our simulation Bars, frequencies of 10000 measured bit strings of the $L = 3$ circuit. Line, exact target.

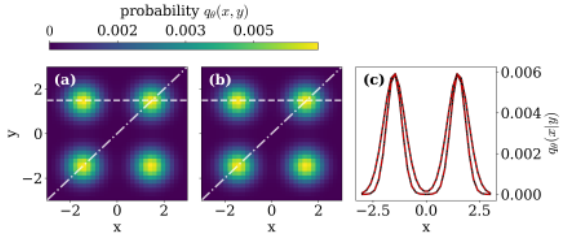
A circuit on a device is used and tested through its samples. The sampled histogram reproduces both peaks, their weights and widths.

Its total variation distance to the target is 0.051, close to 0.055 of the exact model probabilities. The KL divergence from samples is infinite once a populated bin receives no shot, so samples are judged by d_{TV}.

Example from the Literature



The quantum scrambling Born machine fixes each entangling block $\hat{U}_s$, a scrambling unitary that spreads correlations over the register, and trains only one-qubit rotations. Haar-random, random brickwork and fixed-time Hamiltonian scramblers reach similar accuracy. Unmeasured ancillas give mixtures of Born distributions, $q(\mathbf{b}) = \sum_{\mathbf{a}} |\langle \mathbf{b}, \mathbf{a} | \psi \rangle|^2$.



Published result, not our simulation. (a) Target with four Gaussian peaks on two variables, (b) learned distribution, (c) slices along the dashed lines, target black and model red.

M. Płodzień, Phys. Rev. A **114**, 012464 (2026), doi:10.1103/qccv-flmq

9

Combinatorial Optimisation with QAOA



JAGIELLONIAN UNIVERSITY
IN KRAKOW

Quadratic Unconstrained Binary Optimisation

Many decisions are two-way choices made for many items at once, a delivery to one of two vans, a vertex to one of two groups. When the cost depends on pairs of items, the problem is a QUBO,

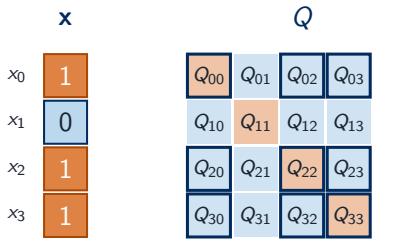
$$\min_{\mathbf{x}\in\{0,1\}^N} C(\mathbf{x}), \qquad C(\mathbf{x}) = \mathbf{x}^T Q\,\mathbf{x} = \sum_{i,j} Q_{ij}\, x_i x_j,$$

with a real symmetric matrix Q. Since $x_i^2 = x_i$, the diagonal Q_{ii} acts as linear costs. Constraints enter as penalty terms.

- Exhaustive search costs 2^N evaluations, about 10^{15} for $N = 50$. QUBO is NP-hard, and large instances are solved by heuristics such as greedy flips or simulated annealing.
- **The answer is a bit string.** Qubit i stores x_i as its bit b_i, and $\hat{Z}_i$ has the eigenvalue $z_i = 1 - 2b_i = \pm 1$. Measuring every qubit in the computational basis returns one candidate $\mathbf{b}$, whose cost a classical computer evaluates at once.

S. Kirkpatrick, C. D. Gelatt and M. P. Vecchi, Science **220**, 671 (1983), doi:10.1126/science.220.4598.671
A. Lucas, Front. Phys. **2**, 5 (2014), doi:10.3389/fphy.2014.00005

The QUBO Matrix



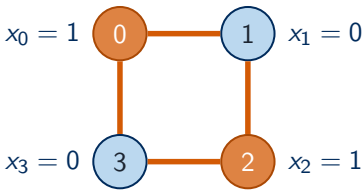
$$C(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x} = \underbrace{\sum_i Q_{ii} x_i}_{\text{linear (diagonal)}} + \underbrace{\sum_{i \neq j} Q_{ij} x_i x_j}_{\text{pairs (off-diagonal)}}$$

only the framed entries contribute

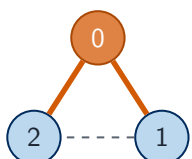
$x_i^2 = x_i$ turns the diagonal into linear costs

The diagonal of Q holds the cost of choosing a single item, the off-diagonal entries the cost of choosing two items together. A candidate $\mathbf{x}$ selects the rows and columns with $x_i = 1$, and the cost sums the entries where they cross.

MaxCut



$\mathbf{x} = 1010$, cut $= 4$



triangle, cut ≤ 2

People are vertices and conflicts are edges. We split them into two rooms, $x_i \in \{0, 1\}$, to separate as many conflicting pairs as possible. An edge (i, j) is *cut* when $x_i \neq x_j$,

$$\mathrm{cut}(\mathbf{x}) = \sum_{(i,j)\in E} \underbrace{(x_i + x_j - 2x_i x_j)}_{\text{1 if cut, 0 if not}},$$

and minimising $C = -\mathrm{cut}$ is a QUBO.

Swapping the rooms never changes a cut, so optimal strings come in complementary pairs, 1010 and 0101 for the square. Odd cycles, such as the triangle, leave one edge uncut.

MaxCut as an Ising Energy

With $x_i = (1 - z_i)/2$ and $z_i = \pm 1$ the eigenvalue of $\hat{Z}_i$, each edge term becomes $x_i + x_j - 2x_i x_j = (1 - z_i z_j)/2$, so

$$\hat{\mathcal{H}}_C = \sum_{(i,j)\in E} \underbrace{\tfrac{1}{2}\hat{Z}_i\hat{Z}_j}_{+\frac{1}{2}\ \text{uncut},\ -\frac{1}{2}\ \text{cut}} \underbrace{-\tfrac{|E|}{2}}_{\text{constant}} .$$

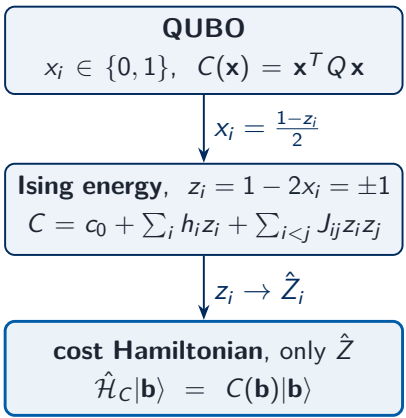
Every basis state $|\mathbf{x}\rangle$ is a candidate split with energy $-\mathrm{cut}(\mathbf{x})$, and the ground states are the maximum cuts. MaxCut is the ground-state problem of an Ising antiferromagnet on the graph.

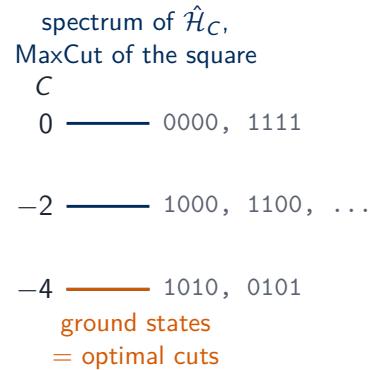
MaxCut is NP-complete. The best known polynomial algorithm guarantees a cut of at least 0.878 of the maximum on every graph.

R. M. Karp, in *Complexity of Computer Computations* (Plenum, 1972), p. 85, doi:10.1007/978-1-4684-2001-2_9

M. X. Goemans and D. P. Williamson, J. ACM **42**, 1115 (1995), doi:10.1145/227683.227684

From a QUBO to a Diagonal Hamiltonian





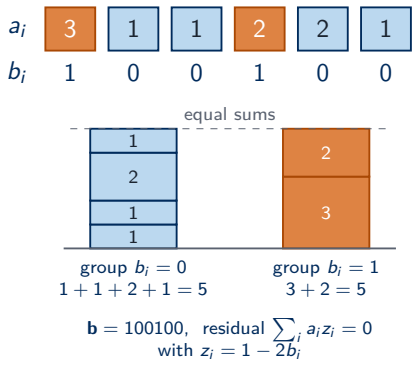
The Ising Coefficients

Inserting $x_i = (1 - z_i)/2$ into $C = \sum_{i,j} Q_{ij} x_i x_j$ and using $z_i^2 = 1$ gives

$$C = \underbrace{\tfrac{1}{4}\sum_{i,j} Q_{ij} + \tfrac{1}{4}\sum_{i} Q_{ii}}_{c_0} \underbrace{- \tfrac{1}{4}\sum_{i} \Big[\sum_{j} (Q_{ij} + Q_{ji})\Big] z_i}_{\sum_i h_i z_i} + \underbrace{\sum_{i<j} \tfrac{1}{4}(Q_{ij} + Q_{ji})\, z_i z_j}_{\sum_{i<j} J_{ij} z_i z_j} .$$

Replacing z_i by $\hat{Z}_i$ gives $\hat{\mathcal{H}}_C = \sum_{i<j} J_{ij} \hat{Z}_i \hat{Z}_j + \sum_i h_i \hat{Z}_i + c_0 \hat{I}$, diagonal in the computational basis. Its ground states are the optimal bit strings.

Number Partitioning



Number partitioning is NP-complete. Every pair is coupled, a complete interaction graph with couplings $\propto a_i a_j$, unlike the sparse MaxCut graph.

Two movers share boxes of weights a_i and want equal loads. Bit b_i puts box i in group 0 or 1, and $z_i = 1 - 2b_i$ turns the difference of the two sums into the residual $\sum_i a_i z_i$,

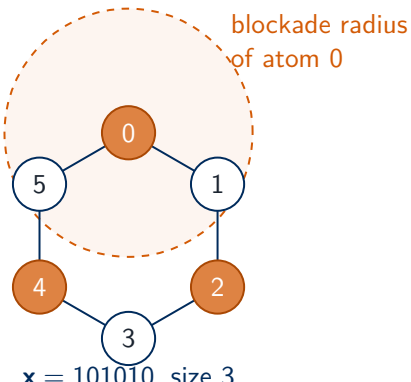
$$C(\mathbf{b}) = \Big(\sum_i a_i z_i\Big)^2 = \underbrace{\sum_i a_i^2}_{z_i^2=1,\ \text{constant}} + \sum_{i<j} \underbrace{2a_i a_j}_{J_{ij}}\, z_i z_j.$$

$$\hat{\mathcal{H}}_C = \sum_{i<j} 2a_i a_j\, \hat{Z}_i \hat{Z}_j + \Big(\sum_i a_i^2\Big)\hat{I},$$

with no fields. A perfect partition has $C = 0$, and perfect partitions are highly degenerate. For our instance $2, 1, 2, 1, 2, 1, 2, 1$, 44 of the 256 bit strings are perfect.

R. M. Karp, in *Complexity of Computer Computations* (Plenum, 1972), p. 85, doi:10.1007/978-1-4684-2001-2_9

Maximum Independent Set



$\mathbf{x} = 101010$, size 3

Transmitters sit at the vertices, and edges join sites that would interfere. We switch on as many as possible with no two neighbours on, an *independent set*, and add the constraint as a penalty with weight B,

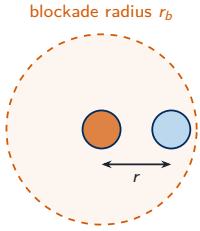
$$C(\mathbf{x}) = \underbrace{-\sum_i x_i}_{\text{chosen vertices}} + \underbrace{B \sum_{(i,j)\in E} x_i x_j}_{\text{violated edges}} .$$

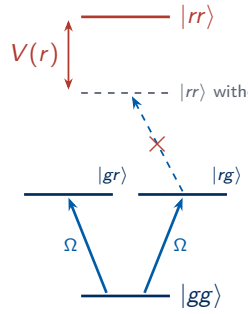
For $B > 1$ dropping an endpoint of a violated edge changes the cost by at most $1 - B < 0$, so every minimiser is a maximum independent set. The ring of six has the maximum sets 101010 and 010101.

The Ising form has $J_{ij} = B/4$ on edges and $h_i = (2 - Bd_i)/4$ for degree d_i. MIS is NP-complete.

R. M. Karp, in *Complexity of Computer Computations* (Plenum, 1972), p. 85, doi:10.1007/978-1-4684-2001-2_9

The Rydberg Blockade





laser in resonance with $|g\rangle \to |r\rangle$
misses $|rr\rangle$ when $V(r) \gg \Omega$,
so at most one atom is excited

A laser tuned to the ground–Rydberg transition excites one atom, but the interaction $V(r)$ shifts the doubly excited state out of resonance. Atoms at the vertices of a graph whose edges join atoms within the blockade radius therefore reach only independent sets.

H. Pichler *et al.*, arXiv:1808.10816 S. Ebadi *et al.*, Science **376**, 1209 (2022), doi:10.1126/science.abo6587

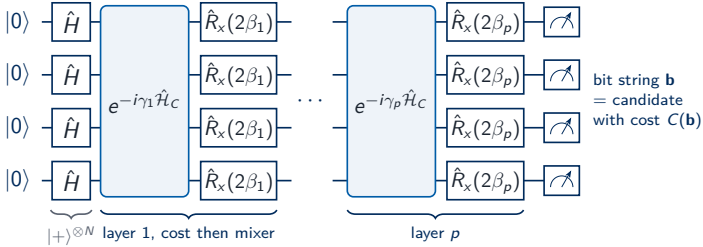
Answers from Measured Bit Strings

A quantum algorithm prepares a state $|\psi\rangle = \sum_{\mathbf{b}} \psi_{\mathbf{b}} |\mathbf{b}\rangle$ and measures all qubits. Each shot returns a candidate $\mathbf{b}$ with probability $|\psi_{\mathbf{b}}|^2$, and we keep the best of M shots. If the optimal strings carry the total probability P_{opt},

$$P_{\mathrm{hit}}(M) = 1 - (1 - P_{\mathrm{opt}})^M, \qquad M_{95} = \left\lceil \frac{\ln 0.05}{\ln(1 - P_{\mathrm{opt}})} \right\rceil \approx \frac{3}{P_{\mathrm{opt}}}.$$

- Uniform random guessing, the state $|+\rangle^{\otimes N}$, gives every string 2^{-N}. A useful state concentrates its weight on low-cost strings.
- In VQE the answer is an energy, a number averaged over many shots. Here the answer is one sampled bit string, whose cost is known exactly once it is measured. The average cost only trains the circuit.

The QAOA Circuit



Initial state $|+\rangle^{\otimes N}$, all candidates equally. **Operations** the cost layer and the mixer, p layers (p plays the role of L), $2p$ angles,

$$|\gamma,\beta\rangle = \prod_{\ell=1}^{p} \underbrace{e^{-i\beta_\ell \sum_i \hat{X}_i}}_{\bigotimes_i \hat{R}_x(2\beta_\ell)} \underbrace{e^{-i\gamma_\ell \hat{\mathcal{H}}_C}}_{\text{phase } e^{-i\gamma_\ell C(\mathbf{b})}} |+\rangle^{\otimes N}, \qquad e^{-i\gamma\hat{\mathcal{H}}_C} \propto \prod_{i<j} \hat{R}_{zz}(2\gamma J_{ij}) \prod_i \hat{R}_z(2\gamma h_i).$$

The cost layer alone leaves all probabilities unchanged and the mixer alone leaves $|+\rangle^{\otimes N}$ unchanged, so only their alternation moves probability.

E. Farhi, J. Goldstone and S. Gutmann, arXiv:1411.4028

Training the QAOA Angles

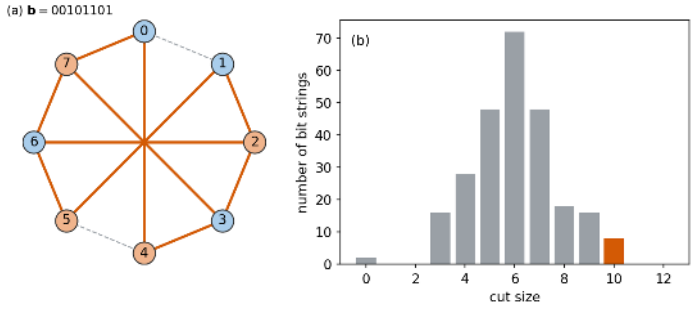
The training signal is the expected cost, the average cost of the measured bit strings,

$$E(\gamma,\beta) = \langle \gamma,\beta|\hat{\mathcal{H}}_C|\gamma,\beta\rangle = \sum_{\mathbf{b}} \underbrace{|\langle \mathbf{b}|\gamma,\beta\rangle|^2}_{\text{probability of } \mathbf{b}}\; C(\mathbf{b}).$$

$\hat{\mathcal{H}}_C$ is diagonal, so E is estimated from the same computational-basis shots that deliver the candidates, with a single measurement setting.

- **Success criteria.** The approximation ratio $r = -E/\mathrm{cut}_{\max}$ for MaxCut, the probability P_{opt} of an optimal string and the shots M_{95} to see one.
- **Gradients.** Every QAOA angle is shared. For MaxCut γ_ℓ enters every $\hat{R}_{zz}(2\gamma_\ell J_{ij})$ and β_ℓ the N gates $\hat{R}_x(2\beta_\ell)$, so the chain rule of the parameter shift counts gate occurrences.
- **Optimisation.** At $p = 1$ a grid over (γ,β) suffices. For $p > 1$ we use exact gradients and Adam from several starts, one of them a linear ramp of the angles.

The MaxCut Instance

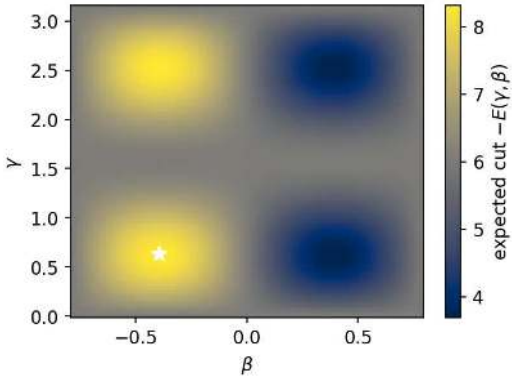


our simulation

A ring of $N = 8$ vertices, each also joined to the opposite vertex, 12 edges. (a) One optimal split, 00101101, cut edges solid, uncut dashed. (b) Number of bit strings for each cut size.

Exhaustive search gives a maximum cut of 10, reached by 8 strings. A random string cuts 6.0 edges on average and is optimal with probability 1/32.

Depth One

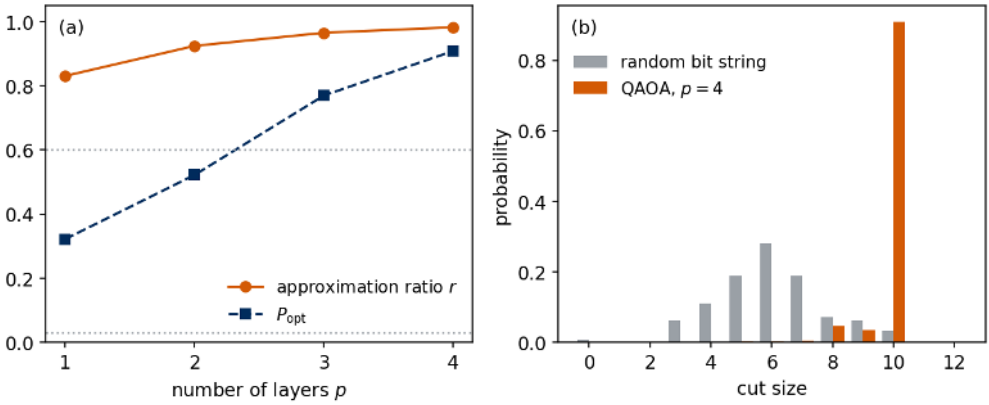


our simulation Expected cut $-E(\gamma, \beta)$ of the $p = 1$ state on a 101×101 grid, star at the best point.

Integer costs make $\hat{U}_C$ periodic in γ with 2π. The shift $\beta \to \beta + \pi/2$ flips every bit, which leaves every cut unchanged, and $(\gamma, \beta) \to (-\gamma, -\beta)$ conjugates the state. The rectangle $0 \le \gamma \le \pi$, $|\beta| \le \pi/4$ contains every value.

At $\gamma = 0$ or $\beta = 0$ the expected cut stays at the random value of six. The best depth-one state cuts 8.31 edges on average.

Deeper Circuits



our simulation

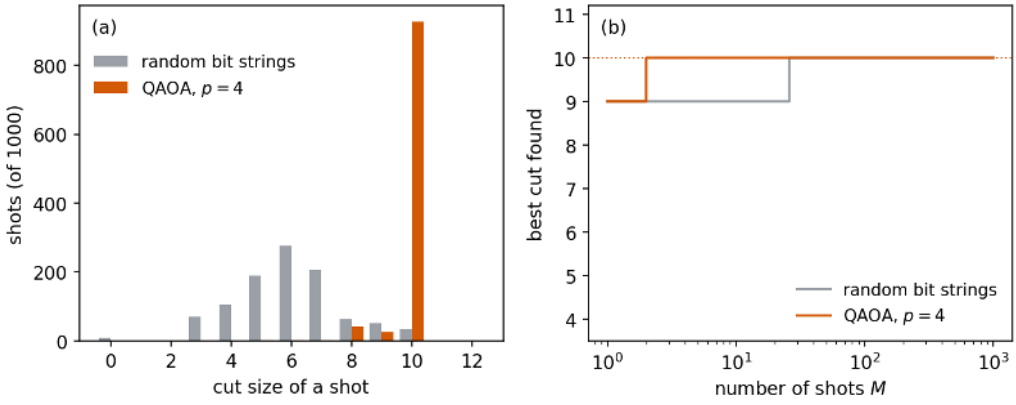
(a) Approximation ratio and P_{opt} against p, random values dotted. (b) Probability of each cut size, random strings (grey) and the trained $p = 4$ state (green).

Quality Against Depth

	r	P_{opt}	M_{95}
random	0.600	0.0312	95
$p = 1$	0.831	0.322	8
$p = 2$	0.925	0.522	5
$p = 3$	0.966	0.770	3
$p = 4$	0.983	0.909	2

Angles trained with exact gradients and Adam from 16 starts per depth. Each layer raises r and P_{opt}. At $p = 4$ the listed result comes from 1 of the 16 starts. On a device one gradient needs $40p$ shifted circuits, since every γ_ℓ enters 12 $\hat{R}_{zz}$ gates and every β_ℓ eight $\hat{R}_x$ gates.

Sampling the Trained Circuit



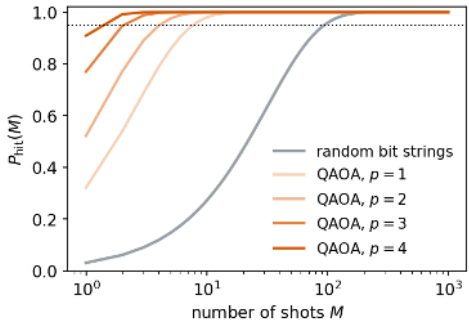
our simulation

(a) Cut sizes of 1000 measured bit strings, trained $p = 4$ state and random strings. (b) Best cut among the first M shots.

Shots, Optimum and Expected Cost

- An optimal string appears in QAOA shot 2 and in shot 26 of random guessing.
- The same shots estimate the expected cost as -9.870 ± 0.016, about 2.5 standard errors from the exact -9.830.
- For eight qubits both sources succeed. For applications what matters is how the required number of shots grows with N, exponentially for random guessing and depending on the problem class and on the depth for QAOA.

Probability of Finding the Optimum

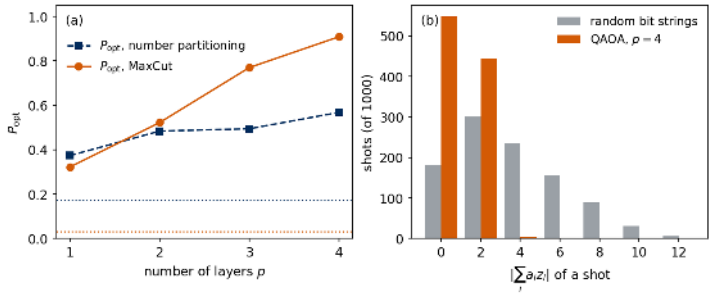


our simulation

$P_{\mathrm{hit}}(M) = 1 - (1 - P_{\mathrm{opt}})^M$ against the number of shots for random bit strings and the trained QAOA states, 95 per cent dotted.

The trained circuits reach a 95 per cent chance of an optimal string within a few shots, random guessing needs about a hundred.

QAOA for Number Partitioning

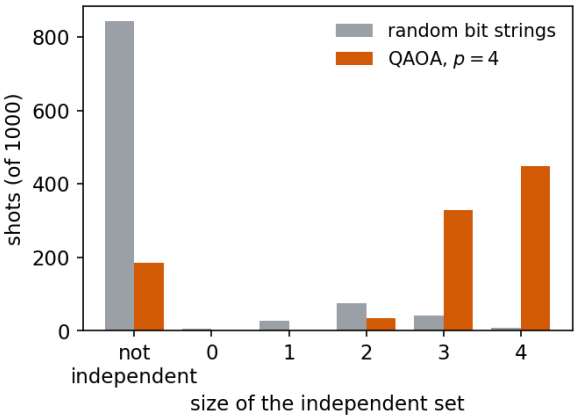


our simulation

Instance $2, 1, 2, 1, 2, 1, 2, 1$. (a) Probability of a perfect partition against p, with MaxCut for comparison. (b) Residuals of 1000 shots, trained $p = 4$ state and random strings.

P_{opt} grows from the random 0.172 to 0.568 at $p = 4$, a smaller gain than for MaxCut, since 44 perfect strings make random guessing already effective.

QAOA for Independent Sets



our simulation

2×4 grid, $B = 2$. Outcomes of 1000 shots, infeasible strings and independent sets by size, random strings and the trained $p = 4$ state.

At $p = 4$ a maximum set has probability 0.456 against 0.0078 at random, and 0.836 of the weight is feasible against 0.160. Infeasible strings are discarded or repaired classically.

QAOA and Adiabatic Evolution

The QAOA circuit is a Hamiltonian-variational ansatz with $\hat{\mathcal{H}}_C$ in place of the chain couplings. It also follows from slow evolution under

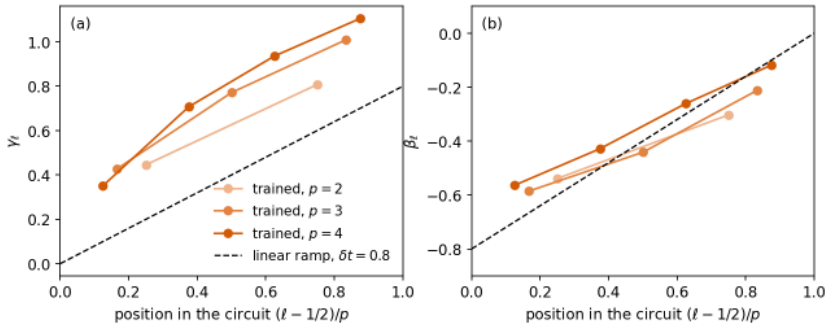
$$\hat{\mathcal{H}}(s) = -(1-s)\sum_i \hat{X}_i + s\,\hat{\mathcal{H}}_C,$$

with s raised slowly from 0 to 1. The ground state of $\hat{\mathcal{H}}(s)$ moves from $|+\rangle^{\otimes N}$ to the optimal bit string. Splitting p steps of length δt into their two commuting groups gives

$$e^{-i\delta t\hat{\mathcal{H}}(s_\ell)} \approx \underbrace{e^{+i(1-s_\ell)\delta t\sum_i \hat{X}_i}}_{\beta_\ell=-(1-s_\ell)\delta t}\ \underbrace{e^{-is_\ell\delta t\hat{\mathcal{H}}_C}}_{\gamma_\ell=s_\ell\delta t},$$

a QAOA layer. As $p\to\infty$ QAOA contains this discretised adiabatic evolution, which reaches the optimum. The trained angles at small p show the same pattern, γ_ℓ growing and $|\beta_\ell|$ shrinking along the circuit.

Trained Angles and the Annealing Ramp



our simulation

Trained (a) γ_ℓ and (b) β_ℓ against the position $(\ell - \frac{1}{2})/p$ in the circuit for $p = 2, 3, 4$, with the linear ramp $\gamma_\ell = s_\ell \delta t$, $\beta_\ell = -(1 - s_\ell)\delta t$, $\delta t = 0.8$ (dashed).

The trained cost angles grow and the mixer angles shrink in magnitude along the circuit, the pattern of a discretised annealing schedule.

Further Problems in QUBO Form

problem	binary variables	cost to minimise
minimum vertex cover	$x_i = 1$ if i is in the cover	$\sum_i x_i + B \sum_{(i,j)\in E} (1 - x_i)(1 - x_j)$
colouring with k colours	$x_{v,c} = 1$ if v has colour c	penalties for one colour per vertex and for equal colours on an edge
travelling salesman, m cities	$x_{c,t} = 1$ if city c is visited at step t	$\sum_{c,c'} d_{cc'} \sum_t x_{c,t} x_{c',t+1}$ plus penalties
knapsack, capacity W	$x_i = 1$ if item i is packed	$-\sum_i v_i x_i + B(\sum_i w_i x_i - W)^2$
Sherrington–Kirkpatrick spin glass	spins $z_i = \pm 1$	$\sum_{i<j} J_{ij} z_i z_j$ with random J_{ij}

$d_{cc'}$ are distances between cities, v_i and w_i values and weights of items. B must exceed the largest gain a violation can bring. The encoding of the constraints sets the number of qubits, Nk for colouring and m^2 for m cities.

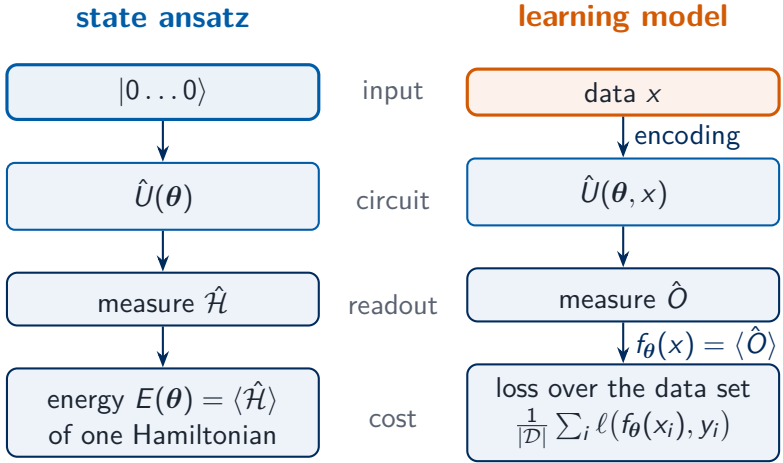
A. Lucas, Front. Phys. **2**, 5 (2014), doi:10.3389/fphy.2014.00005 D. Sherrington and S. Kirkpatrick, PRL **35**, 1792 (1975), doi:10.1103/PhysRevLett.35.1792

10

Circuits as Learning Models

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Two Readings of One Circuit



State Ansätze and Learning Models

Parametrised circuits entered physics as ansätze for many-body ground states. The same circuits were later treated as machine-learning models, first as probability models and then as classifiers.

Quantum circuit learning trains such models with the parameter-shift rule. Circuits have served as classifiers, as feature maps that define quantum kernels, and as models with data re-uploading.

K. Mitarai *et al.*, PRA **98**, 032309 (2018), doi:10.1103/PhysRevA.98.032309 E. Farhi and H. Neven, arXiv:1802.06002 M. Schuld *et al.*, PRA **101**, 032308 (2020), doi:10.1103/PhysRevA.101.032308 V. Havlíček *et al.*, Nature **567**, 209 (2019), doi:10.1038/s41586-019-0980-2 M. Schuld and N. Killoran, PRL **122**, 040504 (2019), doi:10.1103/PhysRevLett.122.040504 A. Pérez-Salinas *et al.*, Quantum **4**, 226 (2020), doi:10.22331/q-2020-02-06-226

11

Summary

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Summary

1. The state of N spins needs 2^N amplitudes. Mean field keeps product states and misses the correlations near the critical point, tensor networks are limited by entanglement and Monte Carlo by the sign problem.
2. The variational principle makes every trial state an upper bound on E_0. A parametrised circuit prepares the trial state on the register, and a classical computer updates its angles.
3. The energy is estimated from bit strings in a few measurement settings with an error proportional to $M^{-1/2}$.

Summary, Continued

4. Along each angle the energy is a cosine, which gives the exact parameter-shift rule. On a device one gradient costs $2\,n_{\mathrm{rot}}\,G\,M$ shots, with n_{rot} parametrised Pauli rotations, G settings and M shots per setting. In simulation automatic differentiation gives it at the cost of a few energies.

5. In our runs (four restarts, 600 Adam steps), ansätze built from the Hamiltonian or restricted to its symmetry sector reached better states with fewer angles than the hardware-efficient ansatz.

6. Energy errors are second order in the state error and entanglement errors first order. In ordered phases a symmetry-broken state has almost the right energy, half the fidelity and no entanglement.

Summary, Concluded

7. Deep random circuits and global costs have exponentially small gradients (barren plateaus), which local costs and shallow, structured circuits avoid.

8. A Born machine models a distribution over bit strings by squared amplitudes. Continuous data are binned, training minimises the KL divergence (the negative log-likelihood up to a constant), and entanglers create the correlations between bits.

9. A QUBO cost becomes a diagonal Ising Hamiltonian whose ground state is the optimal bit string. QAOA alternates cost and mixer evolutions, and the answer is the best measured bit string.

Labs and Reading

Labs

- Lab 069, parametrised circuits and their gradients.
- Lab 070, VQE and the choice of ansatz.
- Lab 070a, VQE phase diagram of the XXZ chain in a field.
- Lab 071, barren plateaus.
- Lab 072, modelling probabilities with a quantum circuit.
- Lab 073, QUBO and QAOA.

Reviews and textbooks

- M. Cerezo *et al.*, Variational quantum algorithms, Nat. Rev. Phys. **3**, 625 (2021), doi:10.1038/s42254-021-00348-9.
- K. Bharti *et al.*, Noisy intermediate-scale quantum algorithms, Rev. Mod. Phys. **94**, 015004 (2022), doi:10.1103/RevModPhys.94.015004.
- M. Larocca *et al.*, Barren plateaus in variational quantum computing, Nat. Rev. Phys. **7**, 174 (2025), doi:10.1038/s42254-025-00813-9.
- U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, Ann. Phys. **326**, 96 (2011), doi:10.1016/j.aop.2010.09.012.
- A. Lucas, Ising formulations of many NP problems, Front. Phys. **2**, 5 (2014), doi:10.3389/fphy.2014.00005.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2010), doi:10.1017/CBO9780511976667.